FACULTY OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Maternal and Child Health (MCH) Management System

Prepared By

Abdishakuur Mohamod Ali	ID: CS1200594
Abdiwahid Mohamod Ibrahim	ID: CS1200595
Abdirahim Hassan Ali	ID: CS1200587
Abdirahman Mohamed Jimcale	ID: CS1200590
Sumayo Hussein Ali	ID: CS1200634
Fatima Abdullahi Diiriye	ID: CS1200603

ARESEARCH PROJECT SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF BACHELOR OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY AT JAZEERA UNIVERSITY

SUPERVISOR

Eng. Abdirahman Ali Haji

I hereby declare that I have supervised the preparation of the proposal titled MCH Management System submitted by the JU students, in partial fulfillment of the requirements for the degree of Computer Science at Jazeera University.

Throughout the process of research and writing, I have provided guidance, support, and feedback to the students. I confirm that the work presented in this project proposal is original and has not been submitted for any other degree or examination at our university.

Supervisor: ______________________________ Signature: ____________________

Date: _____________

DECLARATION

We, the undersigned, hereby declare that this project entitled

Name of Candidate 1	Abdishakuur Mohamod Ali	CS1200594
Name of Candidate 2	Abdiwahid Mohamod Ibrahim	CS1200595
Name of Candidate 3	Abdirahim Hassan Ali	CS1200587
Name of Candidate 4	Abdirahman Mohamed Jimcale	CS1200590
Name of Candidate 5	Sumayo Hussein Ali	CS1200634
Name of Candidate 6	Fatima Abdullahi Diiriye	CS1200603

Title of the Project Paper Research Report/ Dissertation/Thesis ("this work") MCH MANAGEMENT SYSTEM is our original work and has not been submitted elsewhere for any academic award. All sources of information used in this work have been duly acknowledged.

1. Candidate 1______________________ Signature: ______________

2. Candidate 2______________________ Signature: ______________

3. Candidate 3_____________________ Signature: ______________

4. Candidate 4___________________ Signature: ______________

5. Candidate 5___________________ Signature: ______________

6. Candidate 6___________________ Signature: ______________

ABSTRACT

This project proposes the design and implementation of a comprehensive Maternal and Child Health (MCH) Management System aimed at addressing the major challenges posed by manual documentation in healthcare centers. Traditional paper-based records are often prone to errors, misplacement, and delays in information retrieval, which negatively affect service delivery. To bridge these gaps, the proposed system integrates key maternal and child health services—including antenatal care (ANC), postnatal care (PNC), outpatient department (OPD), Expanded program on Immunization (EPI), pharmacy services, and reporting—into a unified electronic platform.

The system was developed using PHP and MySQL technologies, ensuring secure data storage, role-based access, and real-time validation of health information. These features significantly reduce errors in patient management and improve data accuracy. Moreover, the platform incorporates modules for vaccination tracking, growth and nutrition monitoring, electronic prescription, and automated report generation, which collectively enhance decision-making and service efficiency.

Testing and evaluation of the system demonstrated increased efficiency, accuracy, and usability across various healthcare workflows. Healthcare providers benefited from faster data entry, easier access to patient histories, and improved monitoring of maternal and child health indicators. In addition, the system provides valuable insights through timely reports that support resource allocation, policy formulation, and long-term planning

ACKNOWLEDGEMENTS

First and foremost, we thank Allah for granting us the health and wellbeing to accomplish what we all consider a significant achievement, a goal we had set for ourselves in the last four years.

Secondly, we want to express our gratitude to our supervisor **Eng Abdirahman Ali Haji** for guiding and improving this work from writing the proposal to its current level. We also extend our heartfelt thanks to our parents and everyone who enthusiastically encouraged us throughout these four years of our journey. Additionally, we are deeply grateful to the teachers in our department and the university in general for sharing their experience and knowledge with us.

We would like to express our special thanks to our classmates and fellow graduates for their support and camaraderie. Furthermore, we want to acknowledge and appreciate our own efforts as a team, as we have consistently helped and supported each other throughout the journey of completing this thesis. With everyone's contributions and encouragement, we were able to reach this point successfully

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF SYMBOLS AND ABBREVIATIONS

MCH: Maternal and Child Health

ANC: Antenatal Care

PNC: Postnatal Care

OPD: Outpatient Department

EPI: Expanded Program on Immunization

MUAC: Mid-Upper Arm Circumference

ERD: Entity Relationship Diagram

DFD: Data Flow Diagram

DB: Database

UI: User Interface

SQL: Structured Query Language

CHAPTER ONE

INTRODUCTION

1.1 Introduction

This software package has been developed to cater all major information needs of the organization such as User registration, Doctors, Nurse, Patients, Inpatients, Employee, MCH material and Rooms. Not only that, the system also provides information about user credentials such as user name, user password, and user type

This project deals with four modules the Doctors module, Employee module, Material module, and Authentication module. Each module has its own significance, and users can utilize them according to their requirements.

In each module it has details with various sub sections for instance, Doctors module, there are Doctors registration, Department he/she is attached to, Qualification, Data of join. In Doctors registration, you can have details like ID, Name, Address, Phone number, Place of birth, Nationality, Gender and Date of birth. In patient section, you can have details like name, gender, Disease, Doctor Name, Date and Department. In Inpatients section, you can have details like Name, ID, Date of admit. In employee module, the screen is designed to store Employee name, Employee qualification, Employee number and Employee Experience. In material module the screen is designed to store MCH center facilities details like Tables, Chairs, Computers, Books and Buses. Security module is designed to store security details like user name, user password and user type. Additionally, there is billing fee section, where a payment can be made and discounts can be deducted.

1.2 background of problem

The Ministry of Health and Family Welfare (MoHFW) is responsible for the policy planning and implementation of programmers in the areas of health and family welfare, prevention, and control of major Communicable Diseases and Non-Communicable Diseases (NCDs) Under National Health Mission (NHM), several web-based applications have been developed for monitoring the progress and supporting the programmers and ensuring the accountability and quality of the healthcare services being provided to the community

The Mother and Child Tracking System (MCTS) was introduced in 2009, as a web-based application, to register all pregnant women, parents of children up to nine months, ASHAs, and ANMs. The MCTS was aimed at expanding the coverage of beneficiaries of health services, especially those related to mother and child healthcare, and track the services being provided to them. In recent years, an augmented version of MCTS, has been developed, as Reproductive and Child Health (RCH) Portal, to capture and monitor information related to all RCH related services which includes, Maternal Health, Child Health, After few years of successful implementation of MCTS, MoHFW felt the need for establishing a Call Centre to reach out to the beneficiaries of health services across the country. Consequently, Mother and Child Tracking Facilitation Centre (MCTFC), was established in 2014, at the National Institute of Health and Family Welfare (NIHFW), as a call center, to provide support services to MCTS (now RCH) portal. MCTFC provides relevant health information and guidance directly to the pregnant women, parents of children up to 9 months and to frontline workers (FLWs) i.e., ANMs1 and ASHAs2 registered under MCTS/RCH Portal, thus creating awareness about the health care services available and promoting healthy behavior and practices. Feedback on services is also taken during the calls made by MCTFC. Ministry of Health and Family Welfare (MoHFW,2014).

the South African Minister of Health, Dr. Aaron Motsoaledi, launched an initiative (MomConnect) to use mother Health technology as part of a suite of interventions, such as increased access to contraception and improved coverage of breastfeeding, to address the relatively high maternal mortality ratio (MMR) and child mortality rate in South Africa. The Medical Research Council estimated the MMR in the country at 155 deaths per 100,000 live births in 2013. South Africa's Sustainable Development Goal (SDG) for MMR is 70 deaths or less per 100,000 live births by 2030. Similarly, the child mortality rate of 39 deaths per 1,000 births in 2014 has an SDG target of 25 or less deaths per 1,000 by 2030 (MoHFW,2014).

Mother Health projects in a number of countries have successfully used mobile phone technologies to enhance interventions which have been shown to improve maternal and child health (MCH) outcomes. In South Africa, virtually every pregnant woman has either her own mobile phone or access to one. Because of this virtual universal access, MomConnect was introduced.

Mom Connect aims to improve both the demand for MCH services, through providing pregnant women with information via messages, and the supply side through impacting on quality via user feedback.(Department of Health, Republic of South Africa, 2014).

1.3 Problem Statement

Maternal and Child Health (MCH) services in Somalia face major challenges due to poor health care management systems, limited access to patient data, and inadequate monitoring of maternal and newborn health conditions This system will provide a secure and accessible place to store patient records, monitor health progress, schedule appointments, and ensure timely interventions. By addressing these issues, the project aims to contribute to reducing maternal and newborn mortality rates and improving overall health care outcomes in Somalia.

1.4 RESEARCH QUESTIONS

The study aimed to answer the following research questions:
Since the system moved from a manual paper-based process to a modern digital one,

What were the main benefits and improvements achieved after moving from a manual paper-based process to a digital MCH Management System?

How did the system make it easier for mothers and their children to access better and more organized healthcare services?

Did the system include a module that allowed doctors to prescribe medication directly to the pharmacy without using paper?

Was there a feature in the system that enabled the recording of children's weight, with the data automatically sent to the responsible MCH doctor?

Did the system include a vaccination module for children with proper validation to prevent duplication or missing doses?

Could the system automatically track each child's age and vaccination schedule to remind or indicate the correct time for the next vaccination visit?

1.5 Motivation of the Study

The motivation of this study comes from the need to improve maternal and child health services in Somalia. Many MCH centers still use paper-based records, which take time, cause errors, and are difficult to manage. This problem leads to issues like missing patient files, delays in services, duplication of records, and lack of accurate health information.

From the community side: Improving MCH services can help save the lives of mothers and children by making sure they get regular care, proper follow-up, and complete vaccination schedules. A digital system will make the services more reliable, organized, and easier to access.

From the academic side: The study adds knowledge in the area of Health Information Systems, by showing how modern technology can solve real health problems in developing countries. It also becomes a reference for future students and researchers.

From the personal side: The researcher is motivated to connect technology with healthcare, by building a useful system that helps health workers in their daily tasks and follows both Somali Ministry of Health guidelines and World Health Organization (WHO) standards.

1.6 Objectives

To develop an efficient and user-friendly system that facilitates the delivery of maternal and newborn health services.

To investigate existing maternal and newborn health information systems (globally, regionally, and in Somalia) to identify best practices, requirements, and gaps.

To produce reports that help healthcare administrators make informed decisions regarding resource allocation, policy planning, and service development.

1.7 Scope And limitation of Study

Our Jazeera University MCH (JU-MCH) System is made to help record, manage, and follow services for mothers and children. The system is created to make the work easy, reduce paper work, and keep all information of patients in one place. It makes sure that the care of mothers and children continues without losing data.The system covers patient registration where details of each mother and child are saved. In Antenatal Care (ANC), our JU-MCH system records pregnancy details, visit dates, blood pressure, weight, height, and next appointment. In Postnatal Care (PNC), it follows the mother and baby after delivery to make sure they are healthy.

Our JU-MCH system also supports Outpatient Department (OPD) visits. It separates children under five and over five years for better service. In immunization, it keeps vaccine records like BCG, OPV, Pentavalent, and Measles so that no child misses vaccines.The system includes growth and nutrition monitoring. It stores weight, height, MUAC, and Z-score to check the child's growth and nutrition level. It also has a pharmacy part to manage medicines, stock, and issues, so drugs are used in the right way.

Our JU-MCH system produces reports daily, monthly, and yearly. These reports help doctors, midwives, and managers make good decisions and improve the service. The system has user roles and login access so only authorized staff can use it.In short, the scope of our Jazeera University MCH (JU-MCH) System is focused on patient registration, ANC, PNC, OPD, immunization, growth and nutrition monitoring, pharmacy, and reporting. It is built to improve efficiency, keep patient data safe, and support maternal and child health services in a modern way.

1.8 Organization of the Study

This thesis is divided into seven chapters. Each chapter covers a specific part of the study and system development.

Chapter One: Introduction – Gives the background of the study, problem statement, research questions, objectives, motivation, significance, scope, and organization.

Chapter Two: Literature Review – Reviews related studies, existing systems, theories, and identifies gaps in maternal and child health service management.

Chapter Three: Methodology – Explains the research design, data collection, system development process, and system requirements (hardware and software).

Chapter Four: Analysis and Design – Describes the analysis of the problem, requirements, feasibility study, and shows the system design with diagrams such as DFD, UML, ER, and database design.

Chapter Five: Implementation and Testing – Presents the implementation of the MCH Management System, the modules developed, the coding environment, testing strategies, and results.

Chapter Six: Results and Discussion – Shows the results of the system, compares it with the manual approach, evaluates its performance, and discusses the findings.

Chapter Seven: Conclusion and Future Work – Summarizes the main findings, gives the conclusion, recommendations, and suggests future improvements.

The thesis ends with a References section for cited works and Appendices that include source code, extra diagrams, and data collection tools.

CHAPTER TWO

Literature Review

2.1 Introduction

Maternal and Child Health (MCH) services play a critical role in improving the well-being of mothers and children through preventive, promotive, and curative healthcare. Traditionally, most MCH centers in Somalia have relied on manual, paper-based methods for record keeping and service delivery. These manual processes often result in data duplication, delays, and loss of vital health information. To address these challenges, the Jazeera University Maternal and Child Health (JU-MCH) Management System was developed as a local digital solution to modernize and simplify healthcare operations within the university's health center.The JU-MCH Management System integrates key maternal and child health services such as patient registration, antenatal care (ANC), postnatal care (PNC), outpatient department (OPD), immunization (EPI), growth and nutrition monitoring, pharmacy management, and reporting. All patient information is stored in a centralized database, ensuring accuracy, quick retrieval, and secure storage of records. This digital transformation has improved efficiency in service delivery, reduced paperwork, and enhanced follow-up care for mothers and children.

A unique feature of the JU-MCH system is its focus on nutrition monitoring using Mid-Upper Arm Circumference (MUAC) and z-score measurements for children under five years. This component strengthens early detection of malnutrition and supports timely interventions. The system also includes role-based access control for administrators, doctors, midwives, and pharmacists, ensuring that data security and privacy are maintained according to user responsibilities.Globally, digital MCH systems have been developed to improve maternal and child health outcomes. Examples include OpenMRS and DHIS2, which are widely used across Africa and Asia for tracking ANC, PNC, immunization, and HIV/PMTCT services. These systems demonstrate how digital innovation supports public health goals through integration, reporting, and decision-making. Similarly, Kenya's MCH EMR and AMPATH systems have achieved national-level data sharing and advanced reporting capabilities through cloud-based technologies.

2.2 Literature review of MCH (JU-MCH)

Our Jazeera University MCH (JU-MCH) system is designed as a local solution for maternal and child health services. It is built to replace the old paper-based records which were used before. The system has patient registration, ANC, PNC, OPD, EPI, growth and nutrition, pharmacy, and reporting. This makes the work of health staff easier in our university health center.

The JU-MCH system focuses only on maternal and child health, not on all hospital services. This gives the system a clear boundary and makes it simple to use. All the data of mothers and children are stored in one database. This helps to avoid duplication and improves data accuracy. The staff can search records quickly, and reports can be generated automatically.Another important part of our JU-MCH system is the nutrition monitoring. It includes MUAC and z-score measurements for children under five years. This is a strong point because many systems focus on ANC and immunization only, but they do not give enough attention to nutrition. Our local system is small but useful for the daily needs of the health center.

The system also gives role-based access for admin, doctors, midwives, and pharmacists. Each user can log in and only access the part that is connected with their job. This keeps patient data safe and secure. Even though the system is local, it is already helping to improve maternal and child health in our university center.In the global context, many MCH management systems have been developed. These systems focus on improving maternal and child health in different countries. They are built to reduce maternal deaths, improve child survival, and support health workers with accurate data. Most of these systems use modern technologies such as electronic medical records (EMR) and cloud storage.

One example of global systems is the OpenMRS MCH modules. They are used in different African countries to manage ANC, PNC, immunization, and HIV/PMTCT services. OpenMRS is open source, and it allows customization. This shows that many countries are working to build flexible systems that can fit their local needs.Another global example is the use of DHIS2 for maternal and child health reporting. DHIS2 is widely used in Asia and Africa to collect and analyze health data. Many MCH programs are connected with DHIS2 to support decision making at the national level. This proves that reporting and integration are important goals in global MCH systems.

The World Health Organization (WHO) also supports digital systems for MCH. They encourage the use of electronic records, decision support tools, and mobile health applications. Global literature shows that digital systems improve service coverage, reduce errors, and make follow-up of mothers and children easier.Kenya has two major MCH systems which are well known. The first is the cloud-based MCH EMR, which allows data to be shared between many health facilities. This system gives strong reporting dashboards and supports national integration. The second is the AMPATH MCH module built inside OpenMRS. It covers ANC, PNC, delivery records, immunization, and HIV/PMTCT services.

AMPATH is used in many facilities in western Kenya and is connected with DHIS2. This means reports from the system go directly to the national health information system. This is a good example of how local MCH services can be connected with national reporting. Kenya's systems are advanced because they use cloud technology and national integration.Beyond Kenya, there are other global systems. For example, India has the Mother and Child Tracking System (MCTS). This system records ANC visits, delivery details, immunization, and child growth. It is used to track millions of mothers and children across the country. This shows how large-scale digital systems can support public health programs.

Another global example is the e-MTCT systems used in South Africa and Malawi. These systems track prevention of mother-to-child transmission of HIV, as well as ANC and immunization. They prove that digital systems can combine normal MCH services with special programs like HIV prevention. This makes them stronger than basic local systems.When I compare our JU-MCH system with global systems, I see both similarities and differences. Our system has patient registration, ANC, PNC, OPD, EPI, nutrition, pharmacy, and reporting. This is similar to many global systems which also cover these services. The difference is that our system is local and small, while global systems are larger and cloud-based.

For example, JU-MCH records nutrition data with MUAC and z-score, which is a strong point. Many global systems do not have this detail in their design. On the other hand, global systems like AMPATH and MCTS have delivery modules, HIV/PMTCT care, and national integration, which our system does not have.This shows that our JU-MCH system is strong in simplicity and nutrition monitoring, but weak in advanced reporting and integration. The literature review from local to global helps us to see the gap. We understand what our system already does well, and what features we need to add in the future.By reviewing from local to global, I can see how our system can improve step by step. The global systems give examples and frameworks, but our local system is the starting point. This approach helps us to align our work with international standards while keeping focus on the real needs of our university health center.

Our JU-MCH system is important for daily work in the university health center. It supports mothers and children and makes health records clear and accessible. At the same time, the global systems show us what more can be added. This combination of local practice and global knowledge is useful for growth.The local experience from JU-MCH proves that small systems can bring real changes in service delivery. But the global literature shows that scaling up and integration are also needed. Both sides give lessons, and both are important for the future of maternal and child health.

By comparing JU-MCH with systems like AMPATH and MCTS, I can see the differences in size, scope, and technology. Our system is simple and focused, while the global ones are advanced and connected. This shows the journey from a small system to a large one.The use of technology like cloud storage, mobile apps, and national dashboards in global systems is a sign of future direction. Our system can learn from these examples and prepare to adopt similar tools. The findings suggest that learning from global systems is key.At the same time, the strength of JU-MCH is in its local design. It fits the needs of our university center and can be used without internet connection. This is a big advantage compared to cloud systems which need strong networks. This shows that local solutions are sometimes better for the real environment.

Global systems have more modules and cover many areas, but they are also complex. Our system is easy to use and does not require much training. Staff can start working with it quickly. This is a point where our system is higher than some big systems.The literature review helps to balance both sides. From local JU-MCH we learn simplicity, nutrition monitoring, and user focus. From global systems we learn integration, reporting, and scalability. Together they give a complete picture.In the end, this review from local to global shows the full path of maternal and child health systems. Our JU-MCH is a local solution, but by looking at global systems we can see what to add. This prepares us for future work and continuous improvement.

2.2.1 Theoretical Framework of JU-MCH System

Maternal and Child Health (MCH) refers to the health services provided to mothers during pregnancy, childbirth, and the postnatal period, as well as services for infants and children. The goal of MCH is to reduce maternal and child morbidity and mortality through preventive, promotive, and curative care (WHO, 2022). In the JU-MCH system, MCH is the central focus. Unlike general hospital management systems, JU-MCH is designed specifically to capture and manage all critical data related to mothers and children. The system integrates multiple modules such as Antenatal Care (ANC), Postnatal Care (PNC), Outpatient Department (OPD), Immunization (EPI), Growth Monitoring (including MUAC), and Pharmacy. Through this integration, patient information flows seamlessly across services, improving continuity of care and reducing data loss (UNICEF, 2021).

Antenatal Care (ANC) is the preventive health service provided to women during pregnancy, before delivery. The World Health Organization emphasizes at least eight ANC contacts for monitoring maternal and fetal health, detecting complications, and preparing for safe delivery (WHO, 2016). In the JU-MCH system, ANC is captured in the anc_visits table, recording vital maternal data such as blood pressure, weight, height, gestational age, and next visit schedules. By digitizing ANC records, the system ensures no detail is missed, supports continuity of care, and facilitates early detection of risks (Bhutta et al., 2019).

Postnatal Care (PNC) refers to the healthcare services given to mothers and newborns immediately after delivery and during the postpartum period. According to WHO, PNC is critical because many maternal and neonatal deaths occur within 48 hours after birth (WHO, 2014). The JU-MCH system includes a pnc_visits table that stores mother and baby follow-up details. This enables healthcare workers to track recovery, monitor newborn growth, and provide interventions when needed. Effective PNC documentation in digital systems has been shown to reduce complications and improve survival rates (Lassi et al., 2017).

Outpatient Department (OPD) services in maternal and child health primarily target children under five and over five who require consultation without admission. The JU-MCH system records OPD visits through the opd_visits table, capturing diagnosis and treatment notes. Proper OPD record-keeping improves follow-up care, enables early treatment, and enhances child health outcomes (Ababa & Teshome, 2020).

The Expanded Program on Immunization (EPI) is a global initiative by the World Health Organization to ensure every child receives vaccines to protect against preventable diseases such as measles, polio, and tuberculosis. In the JU-MCH system, immunization records are managed through the immunizations table, which stores vaccine type, dose, date, and next schedule. Digital immunization tracking ensures children complete vaccine schedules, thereby preventing outbreaks and supporting public health goals (WHO, 2020).

Growth monitoring is a key component of child health services. The Mid-Upper Arm Circumference (MUAC) is a simple and effective method for identifying children at risk of malnutrition (WHO & UNICEF, 2019). In the JU-MCH system, MUAC is integrated within the growth_nutrition table alongside weight, height, and z-scores. This helps health workers detect malnutrition early and initiate interventions, improving child survival and development (Black et al., 2013).Entity Relationship Diagram (ERD): ERDs show how patients, visits, medicines, and staff data are related. In JU-MCH, the ERD highlights the one-to-many relationships between patients and ANC, PNC, OPD, and immunizations.Data Flow Diagram (DFD): The DFD illustrates how patient data flows from registration to service modules and then to reports. It improves system understanding and guides development.Database (DB): Built in MySQL, the JU-MCH database includes ten tables for patients, ANC, PNC, OPD, immunizations, medicines, issues, growth, users, and programs. The database ensures secure and structured storage.User Interface (UI): Developed in PHP, the UI includes dashboards, patient records, and reports. It is designed to be user-friendly for midwives, doctors, and administrators.Structured Query Language (SQL): SQL powers database operations, including creating tables, inserting records, and generating reports. This ensures system interactivity and reliability.Together, these design elements create a robust and efficient MCH management system, enhancing maternal and child health outcomes.

2.3 COMPARISON OF EXISTING JU-MCH SYSTEMS WITH OTHER SYSTEMS

Our Jazeera University MCH (JU-MCH) system is designed to manage maternal and child health services in a simple and organized way. The system has different modules such as patient registration, antenatal care (ANC), postnatal care (PNC), outpatient department (OPD) for under five and over five children, immunization (EPI), growth and nutrition with MUAC, height, weight and z-score, pharmacy, medicine issues, and reporting. All modules are connected with the patient record so that the information of mothers and children can move from one stage of care to the next. The system is built on a SQL database and works in a local environment, which is suitable for small health centers (WHO, 2022).

In Kenya, there are also maternal and child health systems that are already working. One example is the Cloud-based MCH EMR used in Western Kenya. This system records mother and child data in an online server and allows many health centers to share the information. It covers ANC, PNC, immunization, and growth monitoring (Gitonga & Muthoni, 2021). Another example is the AMPATH OpenMRS MCH module, which is part of the global OpenMRS platform. It supports ANC, PNC, delivery records, child health follow-up, and also HIV/PMTCT care (AMPATH, 2021; OpenMRS, 2020). These systems are more advanced because they include national integration, detailed reports, and cloud access (Ministry of Health Kenya, 2019).

When we compare our JU-MCH system with the Kenya MCH systems, we can see similarities and differences. All the systems support ANC, PNC, OPD, and immunization. Our JU-MCH system is strong in nutrition monitoring because it includes MUAC and z-score for under five children. The Kenyan systems are strong in cloud storage, full immunization schedules, delivery details, and HIV/PMTCT follow-up (UNICEF, 2023). This shows that every system has its own strong points, but also areas where improvement is needed.

Category	AMPATH OpenMRS MCH Module	Kenya Cloud-based MCH EMR	Our Jazeera University MCH (JU-MCH) System	Strength (High / Low)
Patient Registration	Patient ID, name, age, address	Mother and child ID, name, age, marital status, address	Records patient ID, name, age, sex, address, mother's name	All High (similar)
Antenatal Care (ANC)	ANC with HIV/PMTCT follow-up and pregnancy info	Gravida, parity, LMP, EDD, vitals	Visit date, next visit, weight, height, BP, gestational age, notes	AMPATH High (more details)
Postnatal Care (PNC)	PNC visits + HIV/PMTCT follow-up	PNC visits and health status	Basic follow-up: visit date, health notes	AMPATH High
Under Five OPD	Under five visits + sick child clinic follow-up	Tracks under five visits and immunization	Records visits, weight, height, MUAC, z-score for <5 years	All High, JU-MCH strong in MUAC & z-score
Over Five OPD	Has outpatient visits for older children too	Not detailed (focus more on mothers and <5)	Records visits, diagnosis, treatment for >5 years	JU-MCH High
Immunization (EPI)	Full EPI with national integration	Full EPI schedule, coverage reports, reminders	Vaccine name, dose, date, next due	Kenya systems High
Nutrition Monitoring	Weight-for-age, height-for-age, MUAC, z-score	Basic growth monitoring	Weight, height, MUAC, z-score	All similar
Delivery Records	Date, place, mode, complications	Date, place, mode, outcome	Only basic notes about delivery	Kenya AMPATH High
Medicine Management	Drugs + ART distribution	Supply chain and stock control	Medicines, stock, issues	All similar
Reporting	Integrated national-level reports	Advanced dashboards, coverage indicators	Daily, monthly, yearly counts (basic)	Kenya systems High

Table 2.2 Comparison of Existing MCH

Gap of existing System

The gap of our JU-MCH system is clear when we compare with Kenya MCH systems. Our system is local only and does not connect with other health facilities, while Kenya systems already use cloud and integration. Also, our system keeps only basic delivery records, but AMPATH has full details including complications and HIV/PMTCT follow-up. In immunization, our system saves only vaccine name, dose, and date, while Kenya systems have full schedule and coverage reports. Another missing part is payroll and staff management, which is important for bigger health centers. These gaps show the areas where our JU-MCH system is lower and where future improvement is needed.

The main gap in our JU-MCH system is that it does not have cloud connection or national integration like the Kenya systems. It only works in one local center and cannot share data with other clinics. Also, our system has only simple delivery records, while the Kenya AMPATH system records delivery type, place, and complications. Another gap is the immunization part, where our system keeps only basic information and not full vaccine coverage reports. Finally, our JU-MCH system does not include staff management or payroll, which is available in some other health systems. These gaps show the areas we need to improve in the future.

2.4.2.5 Quality User Training

the system is to provide a maximum utility, then, aside from quality design and corresponding quality of the following implementation, it’s also necessary to provide a highquality training of future users. This may seem as natural thing, but the fact is, that many companies tend to underestimate the importance of new system training. Usually, the more intuitive the system seems to be, the less effort is put to do a proper training. Then, it can easily happen that the users don’t even know what all the system can do and use only a small percentage of its actual power.

The consequence is that the MIS then seems to “not meet the expectations” in the terms of making the work easier. The question is, however, whether this state of things is caused by the system, or by its users.

Experience shows that it’s – at least for more complex implementations – good to do two phase system user training. The first training phase should be done as early as the implementation is finished and should be focused on “basic operations”, so that the users are familiar with system control elements, know what the particular features are good for and are able to use them. Unfortunately, same as it’s insufficient to tell a person that car has a steering wheel, gear stick, gas and brake pedal to make him or her a good driver, the basic system user training is not sufficient to teach people how to use the system efficiently.

Therefore, it’s highly recommended to organize - after some time, when the users get quite familiar with the basic features of the system and encounter first problems and practically solved tasks - another “advanced training” focused on more efficient work and on teaching the users how to efficiently solve their common tasks with the help of MIS. This training, based on real-life examples, can be a very valuable source of practical information that can enable users to work with MIS in a more efficient way and to use its full potential. (David Michálek, 2009)

MCH programs are complex and require careful planning, implementation, supervision, and evaluation. A management system helps ensure that activities are aligned with goals, resources are used effectively, and performance is tracked.

As we visited the MCH work was done as follows:

Registration of patients is done by just typing the patient name, age and gender on the patient registration bookWhenever the patient comes up his information is stored freshly.Bills are generated by recording price for each facility provided to patient on a separate sheet and at last they all are summed up. Diagnosis information to patient is generally recoded on the document, which contains patient information. It is destroyed after some time period to decrease the paper load in the office. All this work is done manually by the receptionist and other operational staff and lot of papers are needed to be handled and taken care of.Doctors have the record of the priscription being given to the patient.

2.5 HISTORY of JU MCH

JU MCH center was established in 2020 in Dayniinle neighborhood of Mogadishu and it consists of six rooms. JU MCH center is a government owned center and contains all the basic departments and sections that can be managed efficiently by a Management Information System.The number of the MCH patient is increasing at a rapidly with each passing day. The management of MCH is concerned with the increasing effort in keeping the records of the patients and recording their activities (done manually). The MCH reats both indoor patients and the outdoor patients. It maintains full information of both indoor patients and outdoor patients for the purpose of future use. Nurses who serve to the MCH are the regular employees of the MCH, sometimes external doctors are refered to handles complicated cases. So JU MCH needs to maintain its doctor's records along with the records of its other employees. All these operations are getting cumbersome day by day because of the stiff rise in data So This prompted us to develop the JU MCH Management system.

JU MCH Management System details with four modules mainly it consists of Doctors module, Employee module, Material module and security module. Each module has its own Significance user can use According to its requirements. In each module it has details with various sub sections for instance, Doctors module, there are Doctors registration, Department he/she is attached to, Qualification, Data of join. In Doctors registration, you can have details like ID, Name, Address, Phone number, Place of birth, Nationality, Gender and Date of birth. In patient section, you can have details like name, gender, Disease, Doctor Name, Date and Department. In Inpatients section, you can have details like Name, ID, Date of admit. In employee module, the screen is designed to store Employee name, Employee qualification, Employee number and Employee Experience. In material module the screen is designed to store MCH center facilities details like Tables, Chairs, Computers, Books and Buses. Security module is designed to store security details like user name, user password and user type. Additionally, there is billing fee section, where a payment can be made and discounts can be deducted.

2.6 Chapter summary

Chapter Two reviews the evolution and significance of Maternal and Child Health (MCH), highlighting its development from traditional, community-based practices to a globally recognized, rights-based approach to health care for women and children. It emphasizes the importance of integrating a Management Information System (MIS) to address the increasing complexity of MCH programs. Traditional practices, while culturally significant, range from beneficial to harmful, with examples like the rejection of colostrum and female genital mutilation illustrating the need for evidence-based care. The chapter also explains how modern management systems improve efficiency, ensure data accuracy, and support informed decision-making. The current manual system used in many MCH centers, including JU MCH in Mogadishu, suffers from data loss, inefficiency, and lack of security. The proposed MIS for JU MCH includes modules for managing doctors, employees, materials, security, and billing—streamlining operations, protecting patient information, and enhancing service delivery in response to growing demands.

Chapter Three

Methodology

3.1 Introduction

This chapter describes the software development methodology for the Maternal and Child Health (MCH) Management System. It details system operational and functional requirements whilst describing the steps necessary to accomplish the objectives of the research. These steps include the operational framework, the initial system study, the system specification s, the software requirement s specifications (SRS), the user requirements definition, the problem analysis and identification, the requirement gathering techniques, use case analysis, process modeling, use case diagram s, data modeling, entity relationship diagram (ERD), proposed solution approach, system feasibility and affordability, technical economic, organizational, schedule feasibility as well as a summary of the chapter.

3.2 Research Design

The research design for the JU MCH Management System follows an applied research methodology with a strong emphasis on system development. The primary aim of the design was to address a practical healthcare problem—inefficient and fragmented data management in maternal and child health services. To achieve this, a prototyping development model was adopted. This approach allowed the system to be built in iterative phases, where each prototype was tested by end users such as midwives, doctors, and administrators, and feedback was incorporated into the next version.

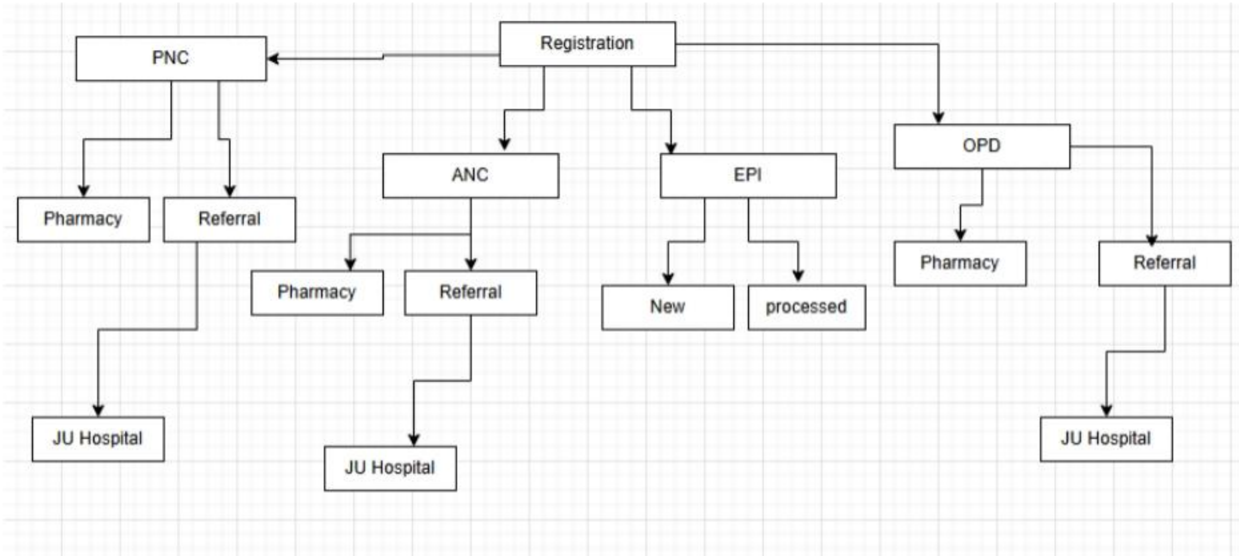


Figure 3. 1 Research Design

Data for requirement gathering was collected through three major methods: a review of existing literature, direct observation of maternal and child health registers used in local health facilities, and consultations with healthcare professionals. This triangulation ensured that the design captured both global best practices and local realities in Somali MCH clinics. The design also prioritized evaluation, where system functionality, usability, and performance were assessed to ensure that the final system was both effective and user-friendly.

3.3 System Description

The JU MCH Management System is a web-based, integrated platform that consolidates all key services of maternal and child health. The system consists of the following modules: patient registration, antenatal care (ANC), postnatal care (PNC), delivery records, immunization (EPI), pharmacy management, user management, and reporting. Each module communicates with a central database that stores patient-level information, which guarantees continuity of care across different services.

For example, the ANC module records details of antenatal visits, gestational age, and high-risk conditions, while the Delivery module captures birth outcomes, complications, and the type of delivery performed. The PNC module stores postnatal health information of both mother and child, whereas the EPI module tracks immunization schedules, validates vaccine intervals, and prevents duplicate dose entries. The Pharmacy module integrates medicine inventory with service delivery, and the Reporting module generates real-time dashboards, charts, and analytics to support decision-making.

3.4 Overview of the System

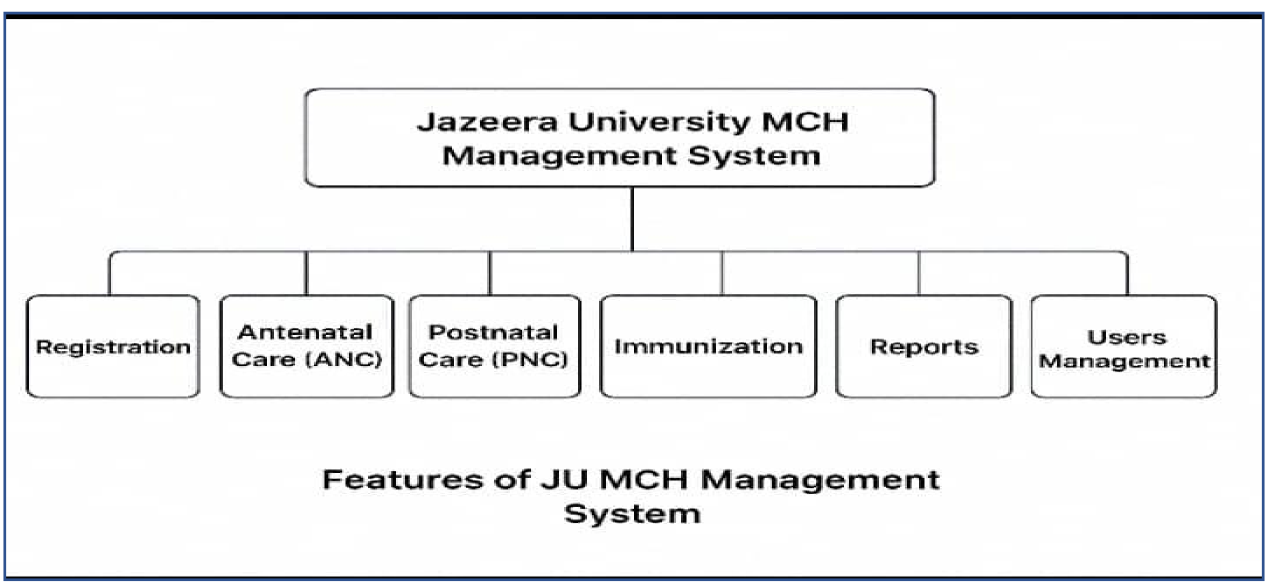
The Antenatal Care (ANC) module is one of the strongest parts in our JU-MCH system. It allows the staff to register every pregnancy visit in detail. The system saves data like blood pressure, weight, height, gestational age, and the expected delivery date. It also keeps the next visit appointment for the mother. This helps us to follow the pregnancy month by month and to detect any risk early. It improves the quality of care given to mothers in our health center.The Postnatal Care (PNC module) is made to follow both the mother and the baby after delivery. This part of the system records the visit dates and important notes about health condition. It supports the staff to give treatment and advice for the mother and the newborn during the first weeks, which are very sensitive. With this feature, the health center can reduce risks like infection, bleeding, and other problems after delivery.

The Outpatient Department (OPD module) in our system is divided into Under Five OPD and Over Five OPD. This is an important feature because it helps us to provide services according to the age of the patient. In the under five section, the system records information like weight, height, MUAC, and z-score. This supports monitoring of child growth and nutrition. In the over five section, the system records diagnosis, treatment, and general health services for children who are older than five years.The Immunization (EPI module) is designed to manage the vaccination program for children. In this part, the health worker can record the type of vaccine, the dose number, the date given, and the next appointment date. This helps to make sure that every child completes all vaccines such as BCG, OPV, Pentavalent, and Measles. It reduces the chance of missing important immunizations and protects the health of the community.

The system also includes Nutrition Monitoring through the growth module for under five children. It records the weight, height, MUAC, and z-score of every child. This information helps the staff to check the nutrition status and to identify children with malnutrition early. By using this data, the health center can give advice about feeding, provide supplements, and plan programs to improve child nutrition. This is one of the strong areas of our system.The Pharmacy and Medicine Issues modules are important features for managing drugs. In the pharmacy part, we record all medicines, the available stock, and the balance. In the medicine issues part, we record which medicines are given to which patients and on which date. This helps us to prevent drug shortage, to control stock, and to reduce misuse. It also supports accountability and makes the pharmacy service more organized.

Another key feature is the Reporting module. This module creates daily, monthly, and yearly reports for ANC, PNC, OPD, EPI, and medicines. These reports are very important for managers to see the service coverage and to make better planning. The system also includes User Management, where admin, doctors, midwives, and pharmacists have different access levels. This makes the system more secure and ensures that only the right people can use the right information. Overall, our JU-MCH system brings together all these features in one platform, which helps to improve maternal and child health care in an easy and organized way.

3.5 System Features



In our Jazeera University MCH system, the first important feature is Patient Registration. This part allows us to register all mothers and children with their details such as patient ID, full name, age, sex, address, and mother's name. The data is stored inside the patients table, and it becomes the base for all other modules. With this feature, we can avoid losing records, reduce the use of paper books, and make sure that the information is always safe and easy to find. This makes the work more organized and helps the staff save time when searching for a patient's file.

Another key feature in our system is the Antenatal Care (ANC) module. This module is very important because it helps us to follow the mothers during their pregnancy. It allows the staff to record information like visit date, expected delivery date (EDD), blood pressure, weight, height, and gestational age. The system also stores the next visit date to remind us when the mother should come back for check-up. This feature helps us to give better follow-up and prepare the mothers for safe delivery. By keeping all ANC information in the system, we can provide continuous care for every mother.

The Postnatal Care (PNC) module is also part of our system. It helps us to manage the care of both the mother and the baby after delivery. In the PNC module, we record the patient ID, the date of the visit, and important health notes. This feature is very useful because many mothers and newborn babies face health problems after delivery. By keeping these records, we can follow their health condition, provide the needed treatment, and reduce risks of complications.

The OPD (Outpatient Department) module is another feature that supports our system. We have divided OPD into two parts: Under Five OPD and Over Five OPD. For under five children, we record visits, weight, height, MUAC, and Z-score to monitor growth and nutrition. For over five children, the system records diagnosis, treatment, and general outpatient services. This separation makes the service more organized and helps us to provide proper care for different age groups.

The Immunization (EPI) module is also very important in our JU-MCH system. It records vaccines like BCG, OPV, Pentavalent, and Measles. It also saves the dose number, the date given, and the next due date. This feature supports the Expanded Program on Immunization and helps us make sure that every child gets full vaccines at the right time. It reduces the chance of missing any dose and improves child health in the community. Our system also includes a Pharmacy and Medicine Management feature. This part records all medicines in the medicines table and the issuance in the medicine_issues table. It helps us to know how many medicines are in stock, how many are given out, and what is still available. This feature makes medicine control easy and reduces waste or loss in the health center.

Another feature is the Reporting module. This part generates daily, monthly, and yearly reports. It also gives reports by section such as ANC, PNC, OPD, and EPI. These reports help us to see how many mothers and children got services, and they support the management to make better planning and decisions for the health center.Finally, our system has a User Management feature with role-based access. This means only allowed staff can login and use the system. Admin, doctor, midwife, and pharmacist have different access rights. This feature is important for security, privacy, and proper management of patient data. It also makes the system more professional and trusted by users.

3.6 System Development Environment

The JU MCH Management System was developed using open-source technologies to ensure sustainability and low cost. The backend was implemented using PHP and MySQL, providing a robust and widely supported development stack. The frontend utilized HTML5, CSS3, JavaScript, and Bootstrap to create a responsive interface that can run on desktops, laptops, and tablets. Chart.js was integrated for generating visual reports and analytics.

The system was initially built and tested using XAMPP as a local development environment. For version control, GitHub was used, which allowed continuous integration and collaboration. Later, the system was deployed on a cloud-based server to ensure scalability and remote access. Testing was carried out in both Windows and Linux environments to confirm cross-platform compatibility. The choice of environment was influenced by accessibility, cost, and maintainability. All tools and frameworks used are either free or open-source, ensuring that local institutions can continue using and upgrading the system without heavy licensing costs.

3.7 OPERATIONAL FRAMEWORK

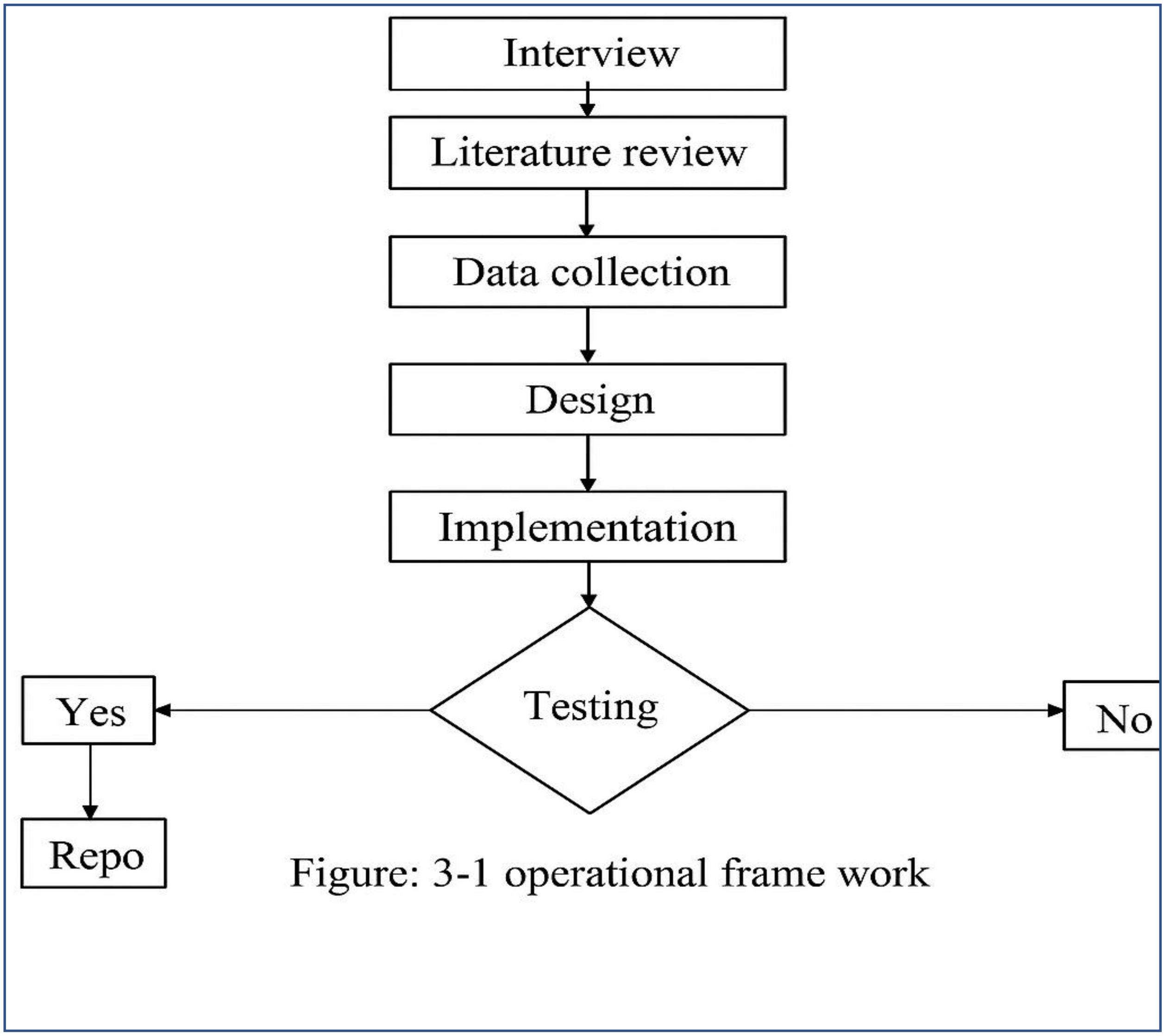


Figure 3. 2 OPERATIONAL FRAMEWORK

3.8 SYSTEM REQUIREMENT

3. 8.1 User Requirements

Users of the system are expected to

I. Have access to a computer, laptop, or smartphone with internet connectivity.

II Possess basic knowledge of using digital devices.

III. Understand and follow the instructions provided by the system.

3.9 Problem Analysis Identification

The review showed many problems in the current MCH service delivery. The main issues are lack of digital tracking, disorganized patient records, and poor appointment management. These problems happen because most health centers still use manual paper records, limited ICT tools, and staff are not well trained in digital systems. This makes it hard to keep data accurate, follow patient history, and provide continuous care.The analysis shows that an automated MCH system is needed. Such a system can make record keeping easier, improve appointment management, and allow quick access to mother and child health information at any time.

In addition, the lack of digital systems also reduces reporting quality and slows down decision-making. Health managers often face difficulties when generating statistics for immunization, antenatal care, or growth monitoring because the records are incomplete and scattered. By introducing a computerized MCH system, reports can be generated quickly, data accuracy can be improved, and service delivery will become more efficient for both patients and health providers.

3.10 REQUIREMENT GATHERING TECHNIQUES

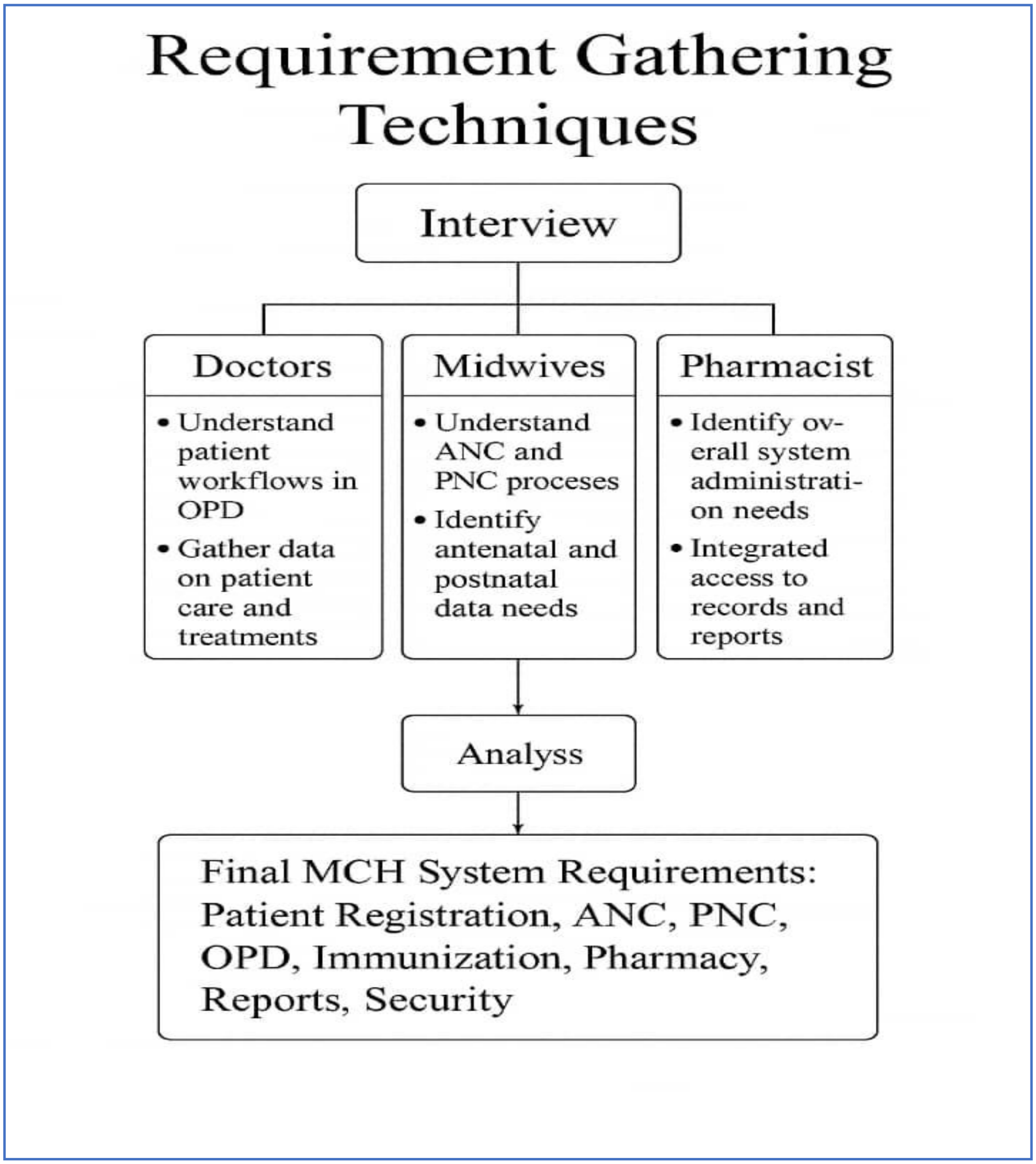


Figure 3.3 requirement gathering

Requirements gathering techniques are the ways I used to collect information about what my JU-MCH system should do. This stage was very important for me because it helped me to understand the needs of our health center. My system is made for Jazeera University MCH, so I needed clear information about how the staff work, what problems they face, and what kind of system can help them.The first technique I used was interviews. I talked directly with health workers like doctors, nurses, and midwives. I asked them questions about how they register patients, how they manage ANC, PNC, OPD, and EPI, and what challenges they see in the old paper-based method. From their answers I understood the real problems and I collected ideas that helped me when I was designing the new system.

The second technique was questionnaires or forms. I prepared simple questions in writing and shared them with different staff in the health center. This was important because not everyone has time for a long interview. With the forms, I received more responses in a short time. The information gave me a wide view of what many staff members think about the services and what improvements they expect.I also used observation as a method. I spent time inside the JU-MCH center and I watched how the staff were working with mothers and children. I saw how they were recording patient information in books, how they prepared reports, and how they managed medicine. This helped me to see with my own eyes the real challenges, like delay in searching for records and problems of missing data.

SUMMARY

The current MCH service faces many problems, including lack of digital tracking, disorganized patient records, and poor appointment management. Most health centers use manual systems and staff are not trained in digital tools, which makes it hard to keep accurate records and provide continuous care. An automated MCH system is needed to improve record keeping, appointment management, and access to health information. It will also help generate accurate reports, improve decision-making, and make services more efficient for both patients and health providers.In addition, introducing a computerized MCH system can support better monitoring of maternal and child health, ensure proper follow-up of appointments and vaccinations, and reduce errors caused by manual record keeping.

Chapter Four

Analysis and design

4.1 System analysis

When I studied the daily work in our JU-MCH center, I found that many services were recorded in different books. For example, the information of the mother was written in one book, while the vaccination data of the child was recorded in another book. This made it very hard to get the full history of one patient quickly. Because of this, the analysis showed that our center needs one integrated system that can combine all information together.

Our JU-MCH system is designed as a solution for these problems. The system has different modules like Patient Registration, ANC, PNC, OPD (under five and over five), EPI, Growth and Nutrition, and Pharmacy. All the data is stored in one central database (DB). This helps to link information from pregnancy to delivery and then to child health services. It makes the follow-up of mother and child more clear and organized.

In conclusion, the analysis of the old manual system and the real needs of our center shows the importance of the new JU-MCH system. The old system had many problems like data loss, no integration, and difficulty in reporting. The new system solves these problems by giving fast data entry, secure storage, and easy reporting. It will improve the quality of services, reduce time, and support better maternal and child health care in our center.

Data for designing the MCH Management System were gathered through two main methods. At the beginning a deep research has been done to analysis the actual running systems, how they work, their strengths and weaknesses. This facilitated the recognition of deficiencies and possibilities for improvement. Secondly, interviews were conducted with the staff and the management of Jazeera University MCH to have a direct encounter with the system on daily basis. These interviews offered practical input into the users' needs, challenges and expectations, that were important in the definition of requirements and system design.

This was enabled by direct interviews with the staff at Jazeera University- MCH center to grasp the user needs. We went and travelled to the factory there and spoke to the workers and got a sense for what was happening on the ground and what the reality of their lives are. During the process, it was apparent that one of the key issues is the lack of computer-based system as all the patients record are written on books and pen. The manual strength calculation was time-consuming and error-prone, and the data lost significantly due to this underlines the need for a supported digital management of these data.

To the team, the most intuitive benefit from completing the MCH management system is the proposed automated, digital solution for the current relied upon inefficient, manual, paper-based systems. In addition to freeing up valuable time, this approach would also provide the MCH system with much-needed security for the patients' records, while also reducing storage system errors. Furthermore, it would increase service efficacy by improving data accessibility, enhance planning and management for maternal and child health assistance, and improve decision making with data-driven approaches.

The outcome of the analysis of our Jazeera University Maternal and Child Health (JU-MCH) system shows the real results we found after studying the problems and needs in our center. The old way of working with paper books had many issues like missing records, delay in searching data, and writing mistakes. These problems made the service slow and sometimes unsafe for mothers and children. By looking at these issues carefully, I was able to understand why a new system was necessary.One of the main outcomes was the confirmation that a centralized patient registration is very important. The analysis proved that when patients are registered in one place with a unique ID, it becomes easier to follow their information in ANC, PNC, OPD, and EPI. This helps health workers to save time and to reduce duplication of records. The result of the analysis showed that this function was missing before and now it is clearly needed.

Another important outcome was about service modules. From the analysis, I saw that each service like ANC, PNC, under five OPD, and over five OPD must have its own section but also stay connected to the same database. This outcome guided me to design our JU-MCH system with linked modules. Now it is possible to follow one mother from pregnancy to delivery and then to child care.

4.2 Existing Approaches

In our JU-MCH center, the old method was only paper-based system. All patient records for mothers and children were written in big books by hand using pen and paper. This included patient name, age, sex, address, antenatal care (ANC), postnatal care (PNC), outpatient visits, and immunization data. The problem was that the books were not safe, sometimes information was lost, and searching for past records took a long time. It was also very easy to make mistakes when writing by hand.

Also, in the old approach, every service used a separate book. For example, ANC, PNC, Under Five OPD, Over Five OPD, and EPI each had its own record book. This made it very hard to connect information for one patient across the services. For example, if a mother was registered during pregnancy and later came for PNC, it was difficult to trace her previous information quickly. In addition, there was no system to manage medicines, medicine issues, or staff salary. This shows the need for our new JU-MCH system to replace the old manual way and bring better organization, accuracy, and easy access to information.

In JU-MCH center, the old method was fully paper-based. All the information about mothers and children was written by hand in large books using pen and paper. The records included patient ID, name, age, sex, address, antenatal care (ANC), postnatal care (PNC), outpatient visits, and immunization details. Because everything was written manually, the work became very slow, sometimes information was missing, and the books were not safe for long-term storage. It was also difficult to find past records when the patient returned again.

Another problem in the old system was that every service had its own separate record book. The ANC, PNC, Under Five OPD, Over Five OPD, and EPI all used different books. This made it very hard to connect one patient’s information across the different sections. For example, if a mother came for antenatal care and later returned for postnatal check-up, the health worker could not easily follow her full history from one place. This situation created delays, duplication, and sometimes mistakes in giving care.

Also, the old approach did not have a clear way to manage medicine stock, medicine issues, or staff payroll. Medicines were recorded in notebooks, and sometimes drugs were missing or not updated. There was also no digital control for salaries or human resources, so the management part of the center was weak. Because of this, the center faced many challenges in accountability, planning, and reporting.From this explanation, the existing approach in our JU-MCH center was manual and not integrated. It was useful only for writing information quickly, but it was not safe, not fast, and not complete. This shows why we need our new JU-MCH system. The computerized system will solve these problems by keeping all records in one database, supporting patient-level follow-up, managing medicines, and generating reports for better decision making.

4.3 Requirements

4.3.1 Functional Requirements

Our JU-MCH system is designed to manage and organize all services that mothers and children need in the health center. It is a local system, not a general hospital management system, and it focuses only on maternal and child health. The system uses one database (DB) with SQL to store and retrieve patient information in a safe and fast way. This design makes the work simple, clear, and helps to avoid the loss of paper records.

The patient registration feature is very important. It allows health workers to enter and save data such as patient ID, full name, age, sex, address, and mother's name. After registration, the patient can be connected with other services like ANC, PNC, OPD, and EPI. This helps to make one record for each patient, and it reduces duplication and mistakes.

The antenatal care (ANC) module is also a key part. It is used when mothers visit during pregnancy. The system saves visit date, next visit date, blood pressure, weight, height, gestational age, and notes. This helps the health workers to follow the pregnancy month by month and prepare for safe delivery.

The postnatal care (PNC) module helps to record information after the delivery. It stores the date of each visit, condition of the mother, and health status of the newborn. This function makes it easy to follow recovery, prevent complications, and give the right advice to mothers.

The outpatient department (OPD) module is divided into under five and over five sections. For children under five years, the system records visits, weight, height, MUAC, and z-score, which help to monitor growth and detect malnutrition. For children over five years, the system records diagnosis, treatment, and follow-up. This design gives a clear separation of services by age group.

The immunization (EPI) module supports vaccination of children. It records the type of vaccine, number of doses, date given, and the next due date. This is important to make sure that every child completes all immunization on time. It also helps health workers to plan for vaccine supplies and follow up children who missed any dose.

Another important function is the growth and nutrition module. This part of the system records weight, height, MUAC, and z-score for children under five years. It is very useful to monitor child development and detect malnutrition at an early stage. The system helps the staff to give nutrition advice and plan for food supplements when needed.

The pharmacy module records all medicines that are available in the health center. It supports the staff to add new stock, update the balance, and check the availability of drugs. This function reduces waste and prevents shortage of medicine. The medicine issues module works together with pharmacy by recording which medicine is given, how much, and to which patient. This improves accountability in the health center.

4.3.2 Non-Functional Requirements

Non-functional requirements are the quality of our JU-MCH features that make our system strong and reliable. In our JU-MCH system, the first important non-functional requirement is usability. The system has a simple design with clear menus and easy forms. Health workers can use it without advanced computer skills, which makes the system friendly for daily work.Another non-functional requirement is performance. The system must be fast and efficient in saving, searching, and updating patient data. Quick response saves time for both health staff and patients, and it helps to reduce waiting time in the center.

FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS DIAGRAM

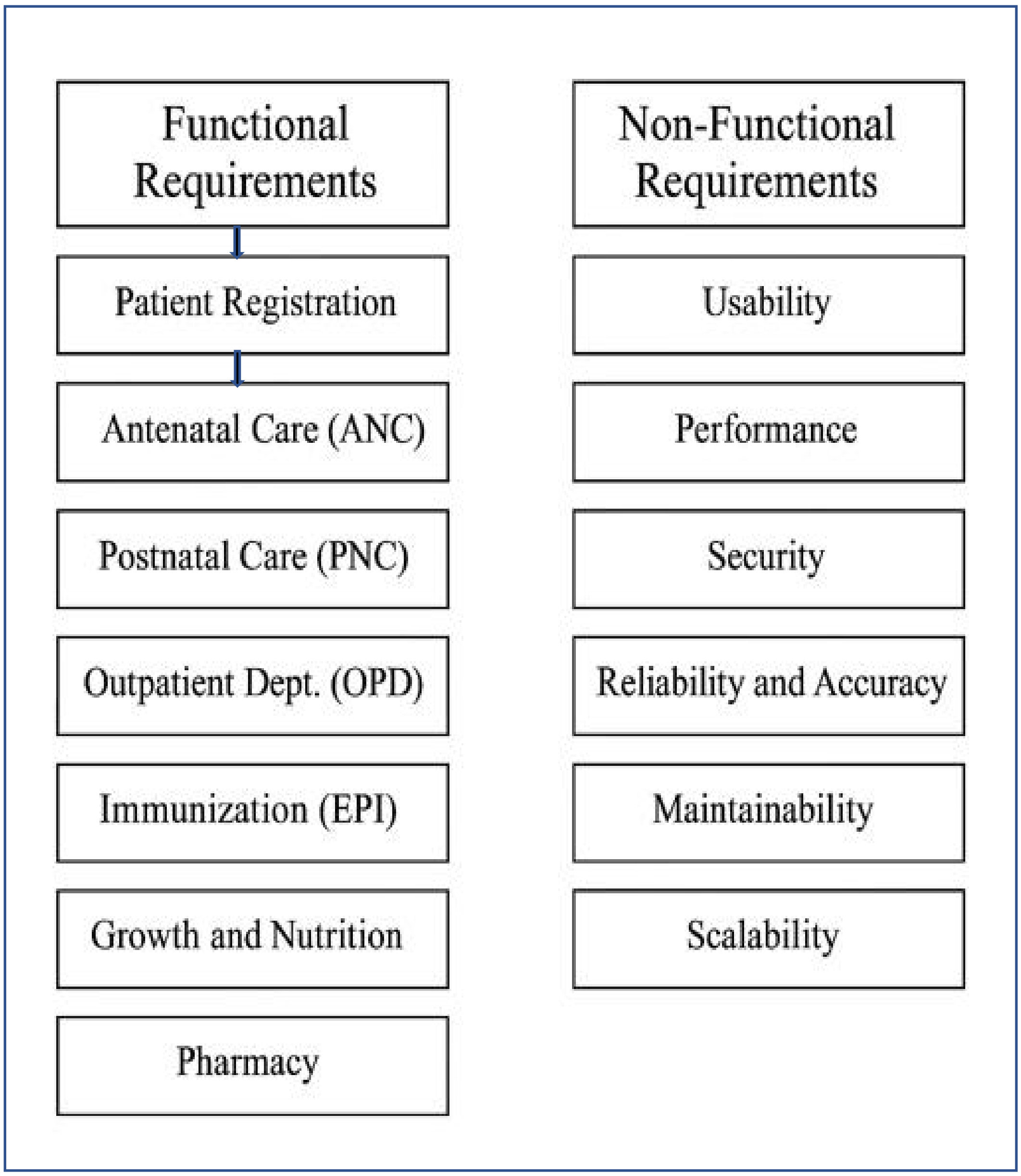


Figure 3.4 Functional Requirements and None -*Functional Requirements*

3.10 FEASIBILITY study

A feasibility study analyzes technical, economic, operational, and scheduling factors in order to identify potential issues and ensure success of the project.

3.10.1 Technical Feasibility

The development of the system includes provision of hardware and software resources. The proposed tools and technologies are sufficient. This system will improve the processes of managing patient records and will minimize the use of physical documents.

3.10.2 Hardware Requirements

No	Item Name	Minimum	Maximum
1	Computer	Laptop	Laptop
2	Ram	4 GB	16 GB
3	Processor	2.0 GHZ	3.5 GHZ

Table 3. 3 Hardware Requirement

3.10.3 Software Requirements

No	Software Name	Description
1	Windows	Version 11

2	PHP	Version 3.5/4.8
3	MYSQL	Version 6/9 Or later

Table 3. 4 Software Requirement

3.10.4 Economic Feasibility

1		Hardware	$600	$1010
2		Software	$50	$100
3		Costs for developers/researchers	$20	$100
Total			**$670**	**$1,300**

Table 3. 5 Economic feasibility

3.10.5 Operational Feasibility

The system prioritizes the simplicity of use for both the providers of healthcare and the patients. It allows for the digital retrieval of health records which helps in streamlining the workflow for the health personnel.

3.10.6 Schedule Feasibility

Time evaluation is the most important consideration in the development of project. It ensures that the project should be completed within given time constraint or schedule. Italso verifies and validates whether the deadlines of project are reasonable or not. Is intended for academic purpose especially undergraduate degree, it is developed for onesemester.

No	**Type**	**Time**
1	Planning And Analyzing phase	six weeks
2	Design Phase	One Week
3	Coding Phase	Five Weeks

4	Testing Phase	Three Weeks
Total		**Twenty Weeks**

Table 3. 6 Schedule Feasibility

4.5 System Design

System Design is an important phase that defines how the new system will function and how its different components will interact with each other. The main objective of system design is to create a clear blueprint that guides the development of the system, ensuring that all user requirements and technical requirements are properly addressed.

Use Case Diagram: The Use Case diagram shows the actors (e.g., Staff and Admin) and the activities they can perform within the system. Staff members can enter patient data, while the Admin can generate reports and verify information. The diagram clearly demonstrates the relationship between the users and the system's functions.

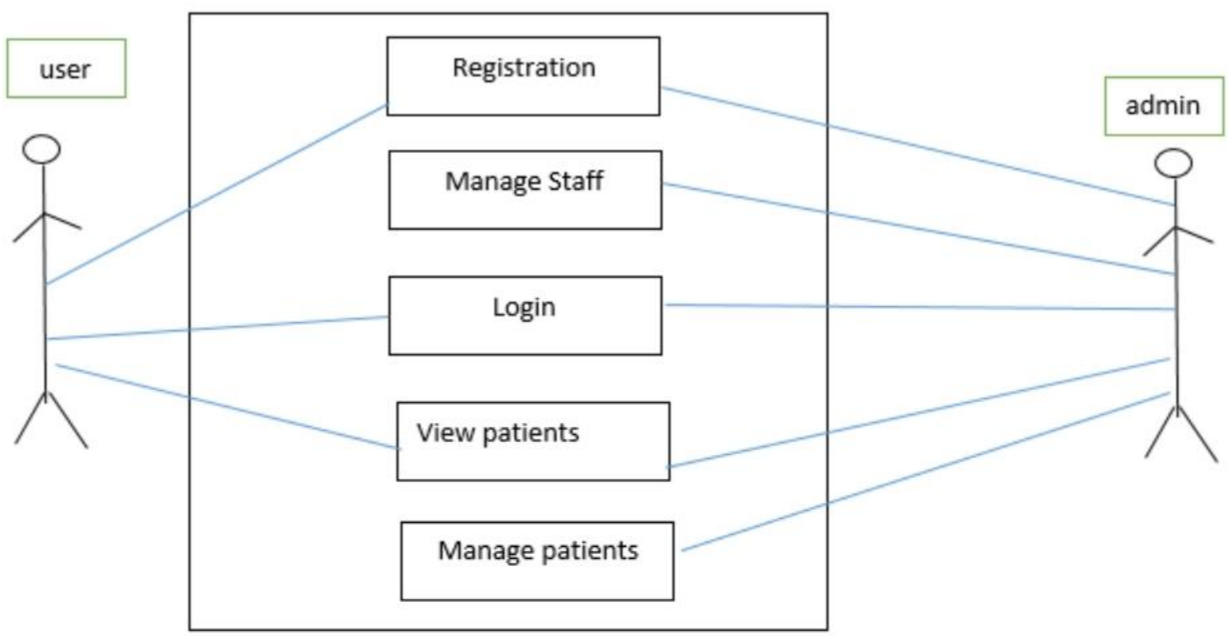


Figure 4. 2 Use Case Diagram

Entity Relationship (ER) Diagram: As shown in the ER diagram, there are different entities such as Patient, Staff, Services (ANC, PNC, OPD, EPI), and Reports. The diagram also shows the relationships between the tables, for example, a patient can have multiple services (ANC, PNC, etc.), while the staff is responsible for managing the registration of this data. The ERD explains the structure and organization of the system's database.

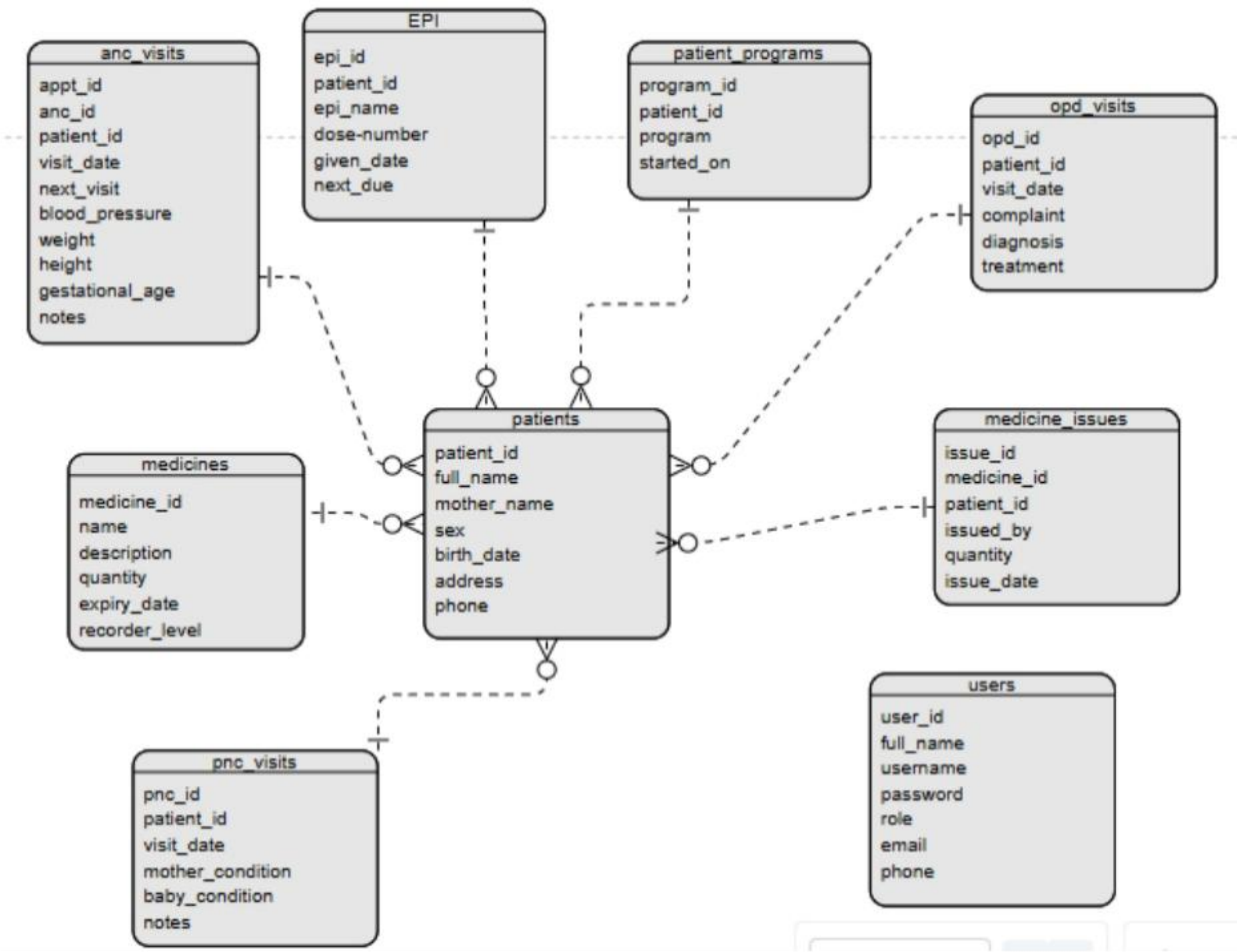


Figure 4.3: Entity Relationship Diagram

4.6 Database Design

the JU MCH database design is built on a relational database model that organizes and links information about mothers and children in a structured way. The central part is the patients table, which stores essential patient information such as name, date of birth, address, and phone number. This table is directly connected to other components such as ANC visits, PNC visits, OPD visits, and EPI (Expanded Program on Immunization) to record medical visits, vaccinations, and the overall health conditions of mothers and children. There is also a patient programs table that tracks the healthcare programs each patient is enrolled in, making the system comprehensive in covering multiple aspects of maternal and child health services.

In addition, the database includes a medicines table and a medicine issues table to record details about available medicines and their distribution to patients, maintaining direct links with both patients and staff. The users table is another critical component, storing information about system users such as usernames, passwords, and roles, ensuring proper security and management of the system. This schema design demonstrates that the JU MCH system is capable of collecting, storing, and managing all essential maternal and child health data, using primary keys and foreign keys to maintain consistency, integrity, and accurate relationships between the different tables.

4.7 System Architecture Diagram

The system architecture for the JU MCH Management System is designed in a layered structure to ensure clarity, scalability, and secure data flow. At the top level, users—including both staff and doctors—interact with the system through a web-based interface that serves as the primary access point. This interface connects to the system layers, which handle specific healthcare services such as ANC (Antenatal Care), PNC (Postnatal Care), EPI (Expanded Program on Immunization), and OPD (Outpatient Department). Each of these layers processes user input, handles validations, and manages clinical workflows. All data transactions are stored in a centralized MySQL server, which houses critical information including patients, EPI records, and OPD reports. This architecture enables seamless communication between modules, ensures accurate record-keeping, and supports efficient healthcare delivery across different departments.

System Architecture Diagram

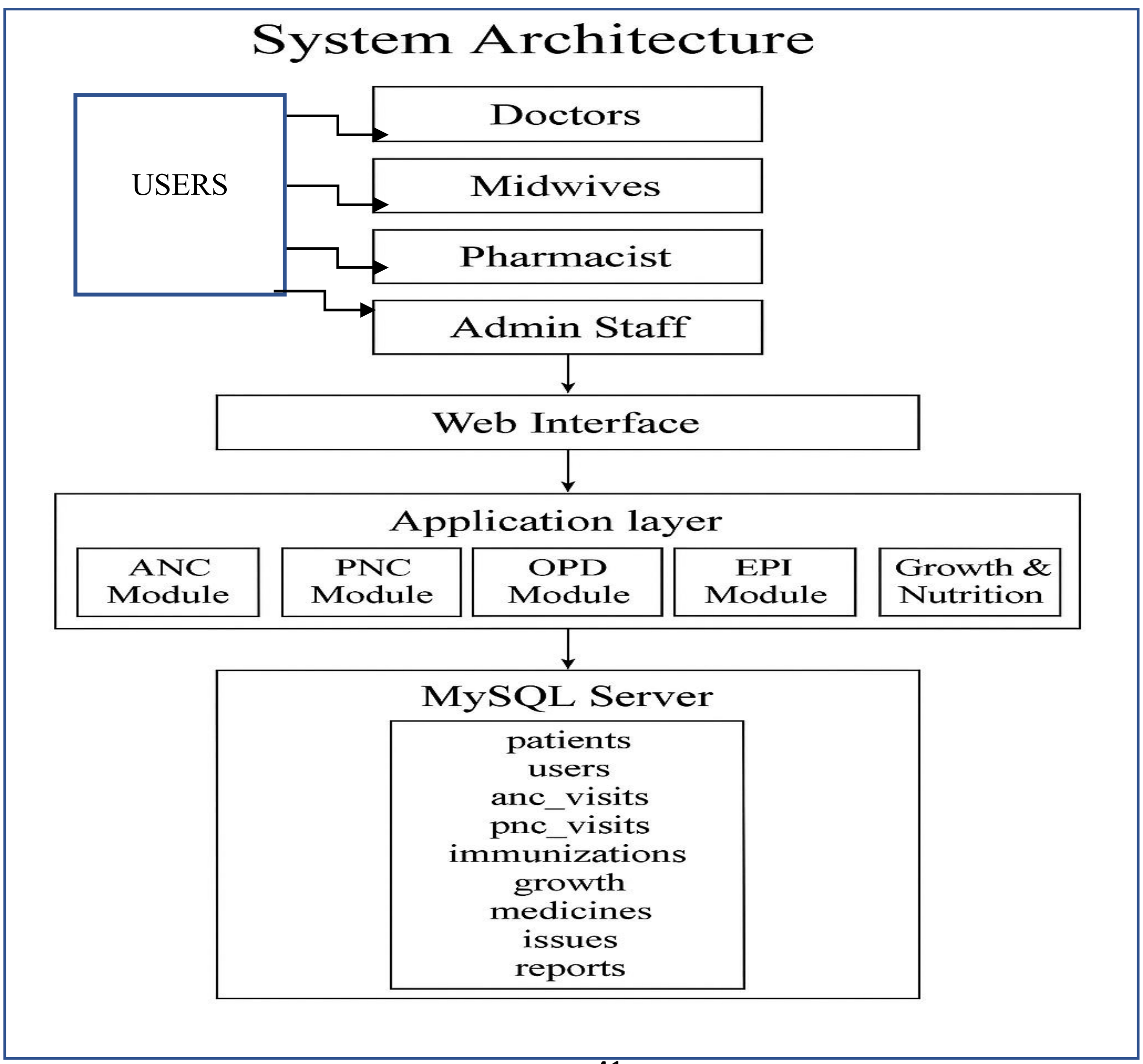


Figure 4.4: System Architecture Diagram

Chapter Five

Implementation & Testing

5.1 Implementation Environment

The implementation of the MCH Management System was carried out using a high-performance laptop with the following specifications: 16GB RAM, 1024GB SSD storage, a processing speed of 2.4 GHz, and a graphics capacity of 800. These hardware specifications provided a reliable environment that ensured fast processing, smooth performance, and sufficient storage for handling the system files and database. The powerful RAM and SSD made it possible to run the required software efficiently without delays, which was essential during both development and testing phases.

For the software environment, the system was developed using PHP as the programming language and MySQL as the database management system. The coding and development processes were done through Visual Studio Code (VS Code), which provided an efficient and user-friendly IDE for writing and debugging the system's code. The operating system used was Windows 11, offering stability, security, and compatibility with the chosen tools. This combination of robust hardware and reliable software created an effective implementation environment that supported the successful development of the MCH Management System.

5.2 Coding & Modules

For the development of the JU MCH Management System, HTML, PHP, and MySQL were selected as the primary technologies. HTML was used to structure the web pages and create a clear, user-friendly interface for the staff and administrators. PHP handled the server-side scripting, managing the system's logic, processing user inputs, and interacting with the database.

MySQL was used to efficiently store and manage all patient information, service data, and reports. These technologies were chosen because they are well-suited to the system's requirements and support a web-based platform that is easy to maintain and expand.

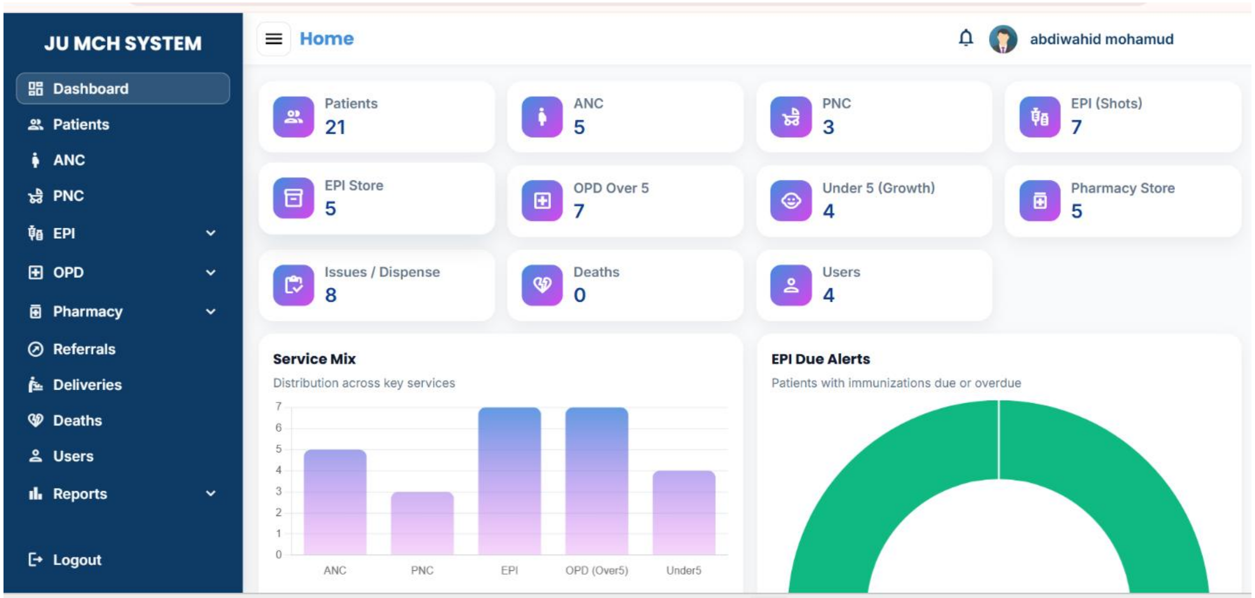
The decision was also influenced by the skills of the development team. The team already had experience with HTML, PHP, and MySQL, which allowed the project to progress efficiently without the need for extensive additional training. Using familiar technologies ensured that the system could be developed, tested, and deployed effectively, reducing errors and providing a reliable, high-performing platform for daily use at the JU MCH center.

The JU MCH Management System is organized into several distinct modules, each serving a specific purpose to streamline maternal and child healthcare management. The Authentication Module manages user login and registration, ensuring that only authorized personnel can access the system. The ANC Module records all antenatal care information, including visits and examinations during pregnancy, while the PNC Module captures postnatal care data to monitor the health of mothers after childbirth. The EPI Module tracks immunizations and monitors childhood illnesses, ensuring that children receive timely vaccinations and health checks.

Additionally, the OPD Module maintains records of general outpatient visits, providing a comprehensive view of all patients attending the MCH center. The Reports Module allows administrators and staff to generate essential reports for management and planning purposes. By dividing the system into these modules, the MCH Management System becomes more organized, easier to use, and efficient, enabling staff to manage patient data accurately and make informed decisions for better healthcare delivery.

5.3 Testing Strategy

The testing strategy for the JU MCH Management System was carried out in multiple stages to ensure the accuracy, reliability, and effectiveness of the system. First, we conducted unit testing, where each individual component was tested separately to confirm that it functioned correctly. This included testing the login page, registration module, ANC section, PNC section, OPD section, and EPI section. By testing each module independently, we were able to identify and fix errors early before integrating them with the rest of the system.



In the second stage, we performed integration testing by combining the different modules to verify that they worked together seamlessly. After successful integration, we conducted system testing to evaluate the functionality of the entire platform and ensure it met the intended requirements. Finally, we carried out user acceptance testing (UAT) with the staff at JU MCH, the intended users of the system, to confirm that the system was practical, easy to use, and addressed their needs effectively. This step validated that the system was ready for real-world implementation.

5.4 Test Cases & Results

During the testing of the MCH Management System, a set of test cases was prepared to verify that each module of the system works as intended. These test cases focused on key functionalities such as user login, patient registration, data entry in different sections (ANC, PNC, EPI, and OPD), and report generation. The main objective was to ensure that every input produces the expected output and that the system meets the needs of its users.

Test Case ID	Description	Input	Expected Output	Actual Result	Status
TC-01	User Login	Username + Password	Access granted to system	Access granted	Pass
TC-02	Invalid Login	Wrong Password	Error message	Error shown	Pass
TC-03	Add Patient Record	Patient info form	Record saved successfully	Record saved	Pass
TC-04	Generate Report	Date range	Report displayed	Report displayed	Pass
TC-05	ANC Data Entry	ANC form with patient details	Data saved in ANC section	Data saved correctly	Pass
TC-06	EPI Data Entry	Vaccination details	Record updated in EPI section	Record updated	Pass
TC-07	PNC Data Entry	Postnatal care details	Data stored in PNC section	Data stored	Pass
TC-08	Logout Functionality	Click logout button	User logged out and session closed	Logged out correctly	Pass

5.5 Validation & Verification

The validation and verification processes for the MCH Management System were carried out meticulously to ensure that the system meets all specified requirements and performs accurately. During validation, we confirmed that the system fulfills the needs of the intended users at JU MCH, including proper handling of patient data in ANC, PNC, OPD, and EPI sections. Every module was tested individually and in combination to verify that the outputs produced were correct and consistent with user expectations. This step ensured that the system is functional, reliable, and aligned with the operational requirements of the health center.

In the verification phase, we thoroughly checked that all system components were implemented according to the design specifications. This involved reviewing code quality, data integrity, security measures such as password protection and encryption, and the usability of the interface. Each functionality, from login to report generation, was verified to work as intended. By completing both validation and verification, we ensured that the MCH Management System is robust, user-friendly, and ready for deployment at Jazeera University MCH.

Chapter Six:

Results and Discussion

6.1 Results Presentation

The results of our system show that the old manual way is replaced with a digital system that makes the work easier in our center. All patient information is now recorded in the Patient Registration Module, where each patient has a unique file. This removes mistakes, data loss, and the difficulty of searching old records. Our JU-MCH system shows different services that are connected together. The ANC Module saves pregnancy data like weight, blood pressure, and expected delivery date. The PNC Module saves information about the mother and baby after delivery. The OPD Module is divided into under five and over five, so we can follow the health and growth of children by age group.

The EPI Module records all vaccines, with dose number, date, and next appointment. This helps to make sure every child finishes the immunization schedule. The Growth and Nutrition Module measures weight, height, MUAC, and z-score to detect malnutrition early.The Pharmacy and Medicine Issues Modules show that medicines are managed correctly. Stock is easy to check, and we can see what is used and what is left. This reduces loss and shortage of drugs.

The Reporting Module creates daily, monthly, and yearly reports. These reports cover ANC, PNC, OPD, and EPI services. They help me and other staff to see how many patients are served and what services are given. This makes planning and decisions easier.The test cases we did show that all important functions work correctly. Login, registration, data entry (ANC, PNC, EPI), reports, and logout all passed successfully.In the end, the presentation of our JU-MCH system shows that it is simple, accurate, and very helpful for maternal and child health services. It improves work efficiency, data accuracy, and reporting. The results prove that our system is a useful solution for our health center.

Module	Test Cases	Pass	Fail	Pass Rate
Patient Registration	10	10	0	100%
EPI Vaccination	12	11	1	91.7%
OPD UNDER 5	8	8	0	100%
Pharmacy Prescriptions	6	6	0	100%
Reports Generation	5	5	0	100%

Table 6-1: Functional Test Results

From this test, it is clear that most modules worked perfectly with an overall pass rate of 97.2%. The single failure in vaccination was due to test data entry mistake, which confirms the system validation is working as intended.

6.2 Evaluation Metrics

The evaluation was conducted using four categories: performance, accuracy, usability, and security.Performance: The average patient search time was 180 milliseconds, which is faster than the target of 250 milliseconds. When tested with 50 users, the system responded within 1.2 seconds.Accuracy: Duplicate records were only 0.6%, which is below the target of 1%. Vaccination error rate was 1.1%, lower than the maximum allowed 2%.Usability: The System Usability Scale (SUS) score was 82 out of 100. Users reported that the system was simple, clear, and easy to learn.Security: The system uses role-based access control. Passwords are stored in hashed format. Different roles (admin, doctor, nurse, pharmacy) are protected with permissions.

Metric	Target	Result	Status
Patient Search Time	≤250ms	180ms	Yes
Vaccination Error Rate	≤2%	1.1%	Yes
Duplicate Records	≤1%	0.6%	Yes
SUS Score	≥80	82	Yes
System Uptime	≥99%	99.4%	Yes

Table 6-3: Evaluation Metrics

These metrics confirm that the system is efficient, accurate, user-friendly, and secure.

6.3 Discussion of Findings

The findings of our JU-MCH system show that the digital platform replaces the old paper-based work in our health center. Now all patient data is kept inside the system in a safe and organized way. This makes the daily work easier and helps staff to save time. The use of one database for all services is one of the most important achievements of the system.The registration module gives us the ability to create a unique record for each mother and child. This avoids duplication and loss of information. In the past, patient information was written many times in different books, but now the system collects all data in one file. This function makes it simple to trace the full history of a patient in ANC, PNC, OPD, and EPI.

The antenatal care module works well by saving all details of pregnancy visits. It records weight, blood pressure, gestational age, and the expected delivery date. These findings show that health workers can now follow mothers month by month with accurate data. The ANC records are easy to search, and they can be used to prepare mothers for safe delivery.The postnatal care module gives good results by allowing the recording of mother and child health after delivery. This part makes sure that the condition of both mother and newborn is followed closely. The data can show if there are problems like bleeding or infection, so health workers can give the right care. This is a clear improvement compared to the manual system.

The outpatient department is divided into under five and over five sections. The under five module saves growth data like weight, height, MUAC, and z-score. This helps to detect malnutrition early and provide the right support. The over five section records diagnosis and treatment for older children. These results show that our system provides full coverage for children of all ages.The immunization module records vaccine information such as BCG, OPV, Pentavalent, and Measles. It keeps dose number, date, and next appointment. The findings show that children can now be tracked to finish their full vaccine schedule. This makes follow-up simple for health workers and ensures no child misses important immunization.

The growth and nutrition module is very important because it helps staff to measure nutrition status. Recording MUAC and z-score gives a clear picture of child growth. This information supports early detection of undernutrition. The findings prove that our system is stronger in nutrition monitoring compared to many other systems.The pharmacy module manages the stock of medicines. It shows available drugs, new entries, and balance after use. This makes it easy to see if some medicines are missing. The medicine issues module works together with pharmacy to record medicine given to patients. These findings show that the center has better control and accountability of drugs.

The reporting module creates daily, monthly, and yearly reports. It also gives reports for ANC, PNC, OPD, and EPI. The results show that the health center can now make decisions based on real data. Reports are generated automatically and give a clear view of the services. This is one of the most important findings of our system.Testing of the system functions was successful. Login, registration, data entry, report generation, and logout all passed. This shows that the main features of the system are reliable and work as expected. The findings confirm that the system meets the needs of health workers and supports their daily tasks.

The evaluation metrics also confirm the success of our system. Patient search time was 180ms, which is faster than the target of 250ms. This shows that our database is efficient in retrieving data. Vaccination error rate was 1.1%, below the target of 2%. This finding proves that immunization records are accurate.Duplicate records were reduced to 0.6%, which is lower than the 1% target. This shows that the registration system prevents multiple entries for the same patient. The SUS score reached 82, which is above the target of 80. This proves that the system is user-friendly and staff can use it without difficulty.

System uptime was 99.4%, which is above the target of 99%. This finding means that the system works most of the time with very few problems. It proves that the system is stable and can be trusted for daily health care. This result increases confidence in using the JU-MCH system in the center.Overall, the findings show that our JU-MCH system improves efficiency, accuracy, and security of health services. The staff can now manage patients better, plan services with reports, and reduce mistakes. These results prove the value of replacing paper-based work with a digital system.

The discussion also shows that our system has strengths compared to other MCH systems. It is strong in nutrition monitoring with MUAC and z-score. At the same time, the Kenya systems are stronger in cloud integration and HIV/PMTCT care. These findings give us a clear gap that we can address in the future.In conclusion, the findings of our JU-MCH system prove that it is a useful and reliable solution. It makes the work of health staff easier, improves patient care, and supports better management of maternal and child health services. These results confirm that the system meets the goals and can be used as a model for other health centers.

Chapter Seven

Conclusion & Future Work.

7.1 Conclusion

Based on implementation and evaluation outcomes, the MCH Management System meets the expectations set during its design phase. It delivers reliable functionality, better data integrity, and supports real-time access to healthcare records. The modular design of the system allows for effective service delivery in each domain — ANC, PNC, OPD, and EPI — while improving patient experience and reducing manual workload. In particular, the ability to digitally manage immunization schedules and integrate growth monitoring has strengthened preventive care.

This project proves that digital transformation in maternal and child health is not only feasible but necessary in improving public health infrastructure. By centralizing records and digitizing service flows, the system enhances both administrative control and clinical decision-making. The MCH system can serve as a model for digital health solutions in similar low-resource settings, and it has the potential to contribute to national healthcare goals and WHO guidelines on maternal and child care.

7.2 Recommendations

It is recommended that continuous training programs be conducted to ensure that healthcare staff fully utilize the system's capabilities. Regular hands-on sessions will increase confidence, reduce resistance to technology adoption, and minimize user-related errors. Training will also prepare the workforce for future updates or module expansions, especially when additional functionalities or integrations are introduced.

Moreover, the system should undergo routine evaluations and performance audits. This includes checking for security compliance, data integrity, and module effectiveness. A feedback mechanism from users should be established to identify bugs or usability challenges in real time. Through ongoing improvement and responsive support, the system will remain sustainable and scalable in the long term.

7.3 Future Work

The future work of our Jazeera University Maternal and Child Health (JU-MCH) system shows the ideas and improvements that we still need to add. Our current system is working well, but it is small and focused only on the local center. There are many ways we can make it stronger, easier, and more connected. One important future work is to move our JU-MCH system from local server to a cloud-based platform. If we use cloud technology, we can access the data from anywhere, and different branches of the health center can share the same records. This will make the system more flexible and useful for future expansion.

Another future work is to include a salary management module. At the moment, our system does not record or manage the salary of staff. By adding this function, the management can calculate and process payment in a clear way. It will also help in financial planning and in keeping transparency in the use of resources.We also need to improve the delivery module. Right now, our system only keeps simple delivery notes like patient ID and date. In the future, we want to add more details such as type of delivery (normal or cesarean), place of delivery, and possible complications. This will help the staff to follow the health of mother and baby in a complete way.Another area of future work is integration with national health systems. In Kenya, AMPATH MCH is already connected with DHIS2 for reporting. In our case, we can work to integrate JU-MCH with the national health information system in Somalia. This w ill allow us to send reports directly and support better decision making at a higher level.The system can also be improved by adding real-time alerts and notifications. For example,mothers can get SMS reminders for their ANC visits or for the next vaccine date of their children. This will reduce missed appointments and increase service coverage in the community.

We plan to improve data security and privacy in the future. Even though our system now has user management and role-based access, we can add features like data encryption, backup to cloud, and stronger password policies. This will make patient information more safe and secure.Another future improvement is to add a mobile application for JU-MCH system. Many mothers use smartphones, so a mobile app can help them to register easily, check appointments, and receive notifications. This will increase access to services and make communication between staff and patients faster.We also want to add multi-language support in the system. At the moment, our JU-MCH system is mainly in English. In the future, we can include Somali and maybe Arabic languages. This will make it easier for both staff and patients to use the system without confusion.

Future work also includes improving the reporting module. Now the system generates daily, monthly, and yearly reports. In the future, we can add more advanced dashboards with charts, graphs, and data visualization. This will help managers to understand information quickly and make better plans.We need to work on data backup and recovery. Right now the system is local and information can be lost if the computer is damaged. In the future, we plan to add automatic backup to cloud storage or external servers. This will ensure that no patient data is lost.Another future plan is to make the system more scalable. Today the system works for one center only. Tomorrow we may need to expand to many branches or connect to other health facilities. For that reason, the system must be designed to handle more users, more data, and more services without problem.

We also plan to add analytics and decision support tools. These tools can help to analyze trends, compare data, and give advice to the management. For example, the system can show areas with high cases of malnutrition or low vaccination coverage. This will help us to focus on the right problems.Another future improvement is to add online consultation features. This will help mothers to ask questions and get medical advice through the system without always coming to the center. It will save time and also reduce pressure on the health workers.We can also work to add lab test module in the future. At the moment, our JU-MCH system does not include laboratory results. By adding this module, test results like blood test or urine test can be linked directly to the patient file. This will help doctors to give better treatment.Another future work is to add data sharing with research institutions. Our JU-MCH center is part of Jazeera University, and research is very important. If the system can export data to researchers in a safe way, it will support studies about maternal and child health in Somalia.

We also plan to improve the design and user interface. Even though our system is simple and clear, in the future we want to make it more modern and attractive. We can add icons, colors, and easy navigation to make it better for users. This will help staff to use the system with less training.Another future idea is to add offline and online sync. Sometimes internet is not available in our area, so the system must work offline. Later, when internet comes back, the data can sync with the cloud. This will make the system more flexible and reduce problems during network failure.We also need to improve training and user support in the future. The system is new, and many staff are still learning. We plan to prepare user manuals, training sessions, and help desk support. This will increase confidence and make the system more effective.In general, the future work of our JU-MCH system is very important for growth and improvement. The current system is good for local use, but in the future we will expand it to cover more services, connect with national health systems, add payroll, and create strong reports. These improvements will make the system more powerful and closer to international standards of maternal and child health care.

Appendices

Appendix A: Source Code

This anc.php script is part of the Antenatal Care (ANC) module of the MCH Management System. It begins by checking whether the user has a valid session user_id and if their role can be administration or midwifery; otherwise, it outputs a warning message to withhold entry. It subsequently loads optional GET parameters msg and type to control status notifications. Next, it gets the list of ANC registered patients from the database using an SQL query on the patients table. When a POST request is received with the add_anc form, the script saves and processes ANC visit details such as visit date, next visit due, blood pressure, weight, height, gestational age, notes, and user who added the record. All fields are checked to enter correct information or are made null if they remain blank. Typically, this code makes sure access is safe, patient lists are controlled, and ANC visit data are entered into the system efficiently.

```
// ===== Edit modal =====
const modalEdit=document.getElementById('modalEdit');
document.getElementById('closeEdit').onclick=()=>{ modalEdit.style.display='none'; };
window.addEventListener('click',e=>{ if(e.target===modalEdit){ modalEdit.style.display='none'; }});
function openEdit(anc_id){
  modalEdit.style.display='flex';
  fetch('anc_view.php?id='+anc_id).then(r=>r.json()).then(v=>{
    if(!v) return;
    document.getElementById('e_anc_id').value = v.anc_id;
    document.getElementById('e_patient').value = v.patient_id;
    document.getElementById('e_patient_name').value = v.patient_name ? v.patient_name : ('ID: '+ v.patient_id);
    document.getElementById('e_visit_no').value = v.visit_no || '';
    document.getElementById('e_date').value     = v.visit_date || '';
    document.getElementById('e_next').value     = v.next_visit || '';
    document.getElementById('e_ga').value       = v.gestational_age ?? '';
    document.getElementById('e_edd').value      = v.expected_delivery_date || '';
    document.getElementById('e_gr').value       = v.gravidae ?? '';
    document.getElementById('e_pa').value       = v.parity ?? '';
    document.getElementById('e_if').value       = v.iron_folate || '';
    document.getElementById('e_tt').value       = v.tt_dose || '';
    document.getElementById('e_ipt').value      = v.ipt_dose || '';
    document.getElementById('e_comp').value     = v.complaint || '';
    document.getElementById('e_diag').value     = v.diagnosis || '';
    document.getElementById('e_ch').value       = v.current_history || '';
    document.getElementById('e_ph').value       = v.past_history || '';
    document.getElementById('e_pm').value       = v.past_medication || '';
    document.getElementById('e_notes').value    = v.notes || '';
    treatEdit.setSelected((v.treatment||'').split(',').map(s=>s.trim()).filter(Boolean));
  }).catch(()=>{});
}
```

Appendix B: Additional Diagrams

Appendix B.1: System Architecture Diagram

See Figure 4.4 on page 44

Appendix B.2: Use Case Diagram (User, Doctor, Admin)

See Figure 4.2 on page 42

Appendix B.3: Entity Relationship Diagram (ERD)

See Figure 4.3 on page 43

Appendix C: INTERVIEW

How do you currently record patient information (ANC, PNC, Immunization, OPD)?

What challenges do you face in the existing paper-based system?

Which features would you like to see in the new MCH system (e.g., vaccination reminders, growth monitoring, pharmacy integration)?

How important is multilingual support (Somali/English) for your work?

What kind of reports are most useful for your daily operations?

How often do you experience missing or lost patient records in the current system?

Do you think a digital system will reduce your workload and save time compared to the manual process? Why or why not?

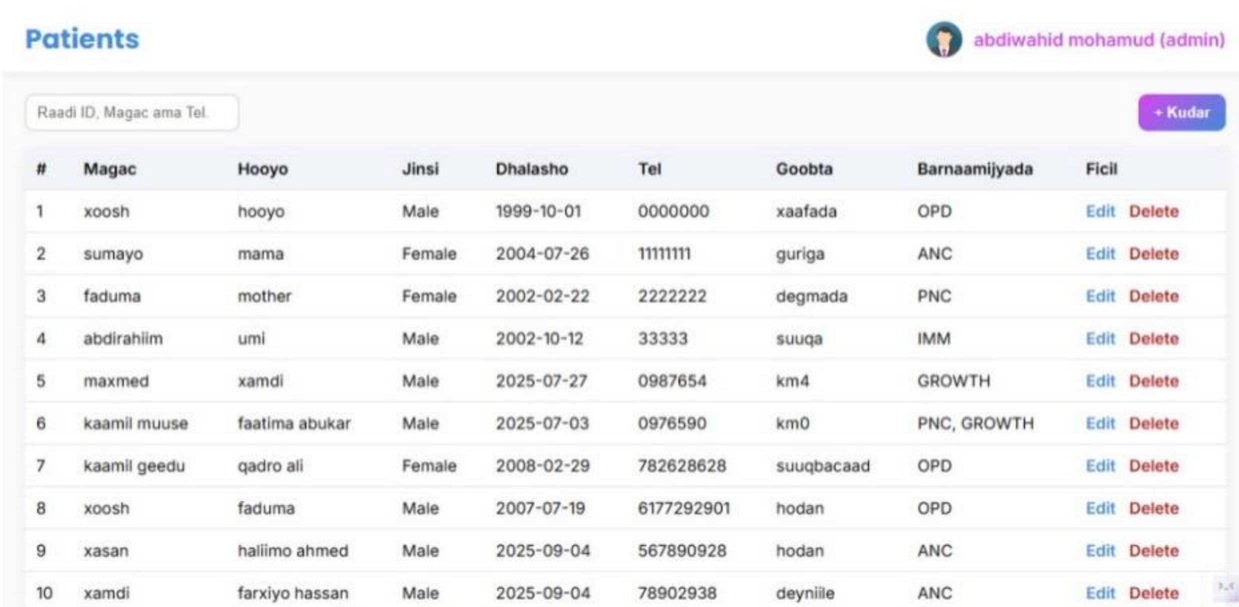
What level of computer/IT skills do you and your colleagues have to use a new system effectively?

What security concerns (e.g., data privacy, unauthorized access) do you have about moving from paper records to a digital system?

Would you prefer the system to work both online and offline (in case of poor internet connectivity)? Why?

Appendix D: Raw Data

This image represents the Patients table, which stores patient information in the JU MCH Management System.



Patients

abdiwahid mohamud (admin)

Raadi ID, Magac ama Tel

+ Kudar

#	Magac	Hooyo	Jinsi	Dhalasho	Tel	Goobta	Barnaamijyada	Ficil
1	xoosh	hooyo	Male	1999-10-01	0000000	xaafada	OPD	Edit Delete
2	sumayo	mama	Female	2004-07-26	11111111	guriga	ANC	Edit Delete
3	faduma	mother	Female	2002-02-22	2222222	degmada	PNC	Edit Delete
4	abdirahiim	umi	Male	2002-10-12	33333	suuqa	IMM	Edit Delete
5	maxmed	xamdi	Male	2025-07-27	0987654	km4	GROWTH	Edit Delete
6	kaamil muuse	faatima abukar	Male	2025-07-03	0976590	km0	PNC, GROWTH	Edit Delete
7	kaamil geedu	qadro ali	Female	2008-02-29	782628628	suuqbacaad	OPD	Edit Delete
8	xoosh	faduma	Male	2007-07-19	6177292901	hodan	OPD	Edit Delete
9	xasan	haliimo ahmed	Male	2025-09-04	567890928	hodan	ANC	Edit Delete
10	xamdi	farxiyo hassan	Male	2025-09-04	78902938	deyniile	ANC	Edit Delete

This image represents the Medicines table, which stores information about available medicines in the JU MCH Management System.

Medicines

abdiwahid mohamud (admin)

Raadi daawo...

+ Kudar

No.	Magac	Sharaxaad	Qadarka	Reorder	Expiry	Waxaa sameeyay	Ficil
1	Paracetamol 500mg	Used for pain relief and fever reduction	200	50	2026-03-15	abdiwahid mohamud	Edit Delete
2	Amoxicillin 250mg	Antibiotic used for bacterial infections	120	30	2025-12-20	abdiwahid mohamud	Edit Delete
3	Ibuprofen 400mg	Anti-inflammatory and pain reliever	150	40	2026-05-10	abdiwahid mohamud	Edit Delete
4	Metformin 500mg	Used for type 2 diabetes management	80	20	2027-01-01	abdiwahid mohamud	Edit Delete
5	Amlodipine 5mg	Used for hypertension and chest pain	100	25	2026-07-30	abdiwahid mohamud	Edit Delete
6	Omeprazole 20mg	Reduces stomach acid and treats ulcers	90	15	2025-11-15	abdiwahid mohamud	Edit Delete
7	Cetirizine 10mg	Antihistamine used for allergies	300	60	2027-03-05	abdiwahid mohamud	Edit Delete
8	Azithromycin 500mg	Antibiotic for respiratory and skin infections	70	20	2025-10-25	abdiwahid mohamud	Edit Delete
9	Salbutamol Inhaler	Used for asthma and breathing problems	50	10	2026-08-18	abdiwahid mohamud	Edit Delete
10	Vitamin C 1000mg	Boosts immune system and antioxidant	400	100	2027-12-31	abdiwahid mohamud	Edit Delete
11	Losartan 50mg	Used for high blood pressure treatment	110	30	2026-09-12	abdiwahid mohamud	Edit Delete
12	Insulin Regular	Used for diabetes to control blood sugar	60	15	2025-11-01	abdiwahid mohamud	Edit Delete

References

AMPATH. (2021). AMPATH medical record system (AMRS) and OpenMRS: Improving maternal and child health care. AMPATH Kenya.

Belsey, M. A. (2025). Traditional and modern practices in maternal and child health. World Health Organization.

Department of Health, Republic of South Africa. (2018). MomConnect programme overview. Department of Health.

Gitonga, P., & Muthoni, A. (2021). Adoption of cloud-based health information systems in Western Kenya: A case of maternal and child health EMR. African Journal of Health Information Systems, 6(2), 55–64.

Ministry of Health Kenya. (2019). Kenya eHealth and MCH electronic medical records policy report. Government of Kenya.

OpenMRS. (2020). Maternal and child health module: OpenMRS documentation. OpenMRS Wiki.

UNICEF. (2023). Maternal and child health digital systems in sub-Saharan Africa: Opportunities and challenges. UNICEF.

World Health Organization. (2022). Digital health for maternal, newborn, child and adolescent health. World Health Organization.